

Lecture Notes

on

PHARMACOLOGY

DRUGS, MECHANISMS & THERAPEUTICS

2nd TERM



Lecture Classes by

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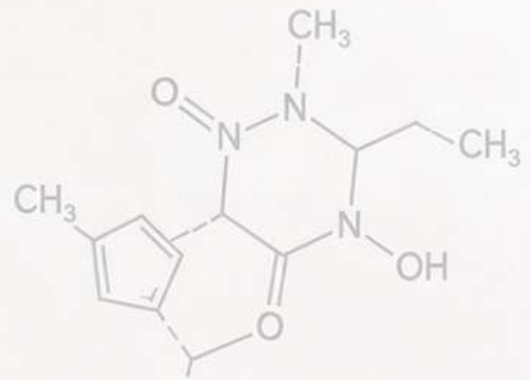
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PHARMACOLOGY • 2ND TERM • LECTURE NOTES

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Lipid Lowering Drugs / Agents

Normal Level

- **Total cholesterol:** < 200 mg/dL
 - **LDL:** < 130 mg/dL
 - **HDL:**
 - > 40 mg/dL (male)
 - > 50 mg/dL (female)
- [HDL is better than LDL]**
- **TG / Triglycerides:** < 150 mg/dL

Hyperlipidemia

- **Total cholesterol:** > 240 mg/dL
- **LDL:** > 160 mg/dL
- **HDL:** < 60 mg/dL
- **TG / Triglycerides:** > 200 mg/dL

Types of Hyperlipidemia

(i) Primary

Cause / predisposing factor:

- Genetic and environmental factor.

(ii) Secondary

Cause:

- (i) Diabetes mellitus
- (ii) Hypothyroidism
- (iii) Primary biliary cirrhosis
- (iv) Excessive alcohol intake.

Risk factor of hyperlipidemia:

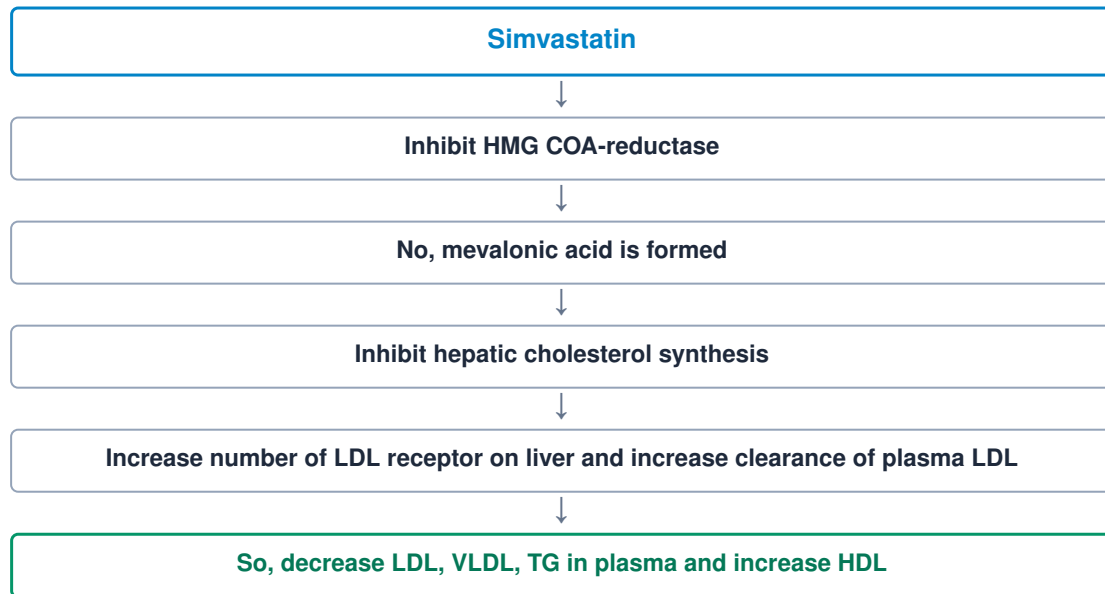
- (i) Smoking
- (ii) Lack of exercise

Classification of Drugs

No.	Class / Mechanism	Drug Examples
(i)	HMG Co-A reductase inhibitor (Statin group of drugs)	Simvastatin, Atorvastatin, Rosuvastatin, Lovastatin, Fluvastatin.
(ii)	Fibrin acid derivatives:	Fenofibrate, Bezafibrate, Gemfibrozil.
(iii)	Bile acid binding resin / Bile acid sequestrants:	Cholestyramine, Colestipol.
(iv)	Inhibition of hepatic production of lipid:	Nicotinic acid.
(v)	Inhibition of intestinal cholesterol absorption:	Ezetimibe

Simvastatin

Mechanism of Action



Clinical Profile

Indication:

- (i) Hyperlipidemia
- (ii) Atherosclerosis
- (iii) Unstable angina

Adverse effect:

- Headache, myalgia.
- GIT - nausea, GIT upset, kidney and liver damage, increase blood sugar level.

Contraindication:

- (i) Liver damage
- (ii) Renal impairment
- (iii) Hypersensitivity
- (iv) Pregnancy
- (v) Lactation
- (vi) Children

Anti-Microbial Agents & Penicillins

1. Anti-Microbial Agents

Definition: The agents that are used against microbes are called antimicrobial agents.

They can kill or inhibit the growth / multiplication of micro-organisms.

Micro-organisms include: Bacteria, Virus, Fungus, Helminth, etc.

★ Antibiotic

Antibiotics are the substances produced by micro-organisms and also produced synthetically that suppress the growth and multiplication of other micro-organisms or kill them without affecting the host tissue.

Chemotherapeutic Agent

Drugs that are produced synthetically which suppress the growth or kill the micro-organisms.

2. Classification of Antibiotics

(I) According to Source:

- 1. Natural: **Benzyl Penicillin**
- 2. Semi-synthetic: **Amoxicillin**
- 3. Synthetic: **Sulfonamides**, **Ciprofloxacin**

(II) According to the Mode of Action:

- 1. **Bacteriostatic:** Antibiotics that suppress the growth of micro-organisms without killing them.
Examples: **Tetracycline**, **Macrolides**, **Chloramphenicol**, **Sulphonamides**, etc.
- 2. **Bactericidal:** Antibiotics that kill micro-organisms.
Examples: **Penicillin**, **Cephalosporin**, **Aminoglycosides**, etc.

(III) According to Spectrum Activity:

- 1. **Narrow Spectrum:** Antibiotics act against either Gram (+ve) or Gram (-ve) micro-organisms.
Examples: **Flucloxacillin**, **Benzyl Penicillin**, **Erythromycin**, etc.
- 2. **Broad Spectrum:** Antibiotics act against both Gram (+ve) and Gram (-ve) micro-organisms.
Examples: **Tetracycline**, **Amoxicillin**, **Ampicillin**, **Cephalosporin**, etc.
- 3. **Extended Spectrum:** Broad spectrum antibiotics with more Gram (-ve) coverage.
Examples: **Carbenicillin**, **Piperacillin**, etc.

(IV) According to Mechanism of Action:

- 1. Inhibit cell wall synthesis:

Examples: **Penicillin**, **Cephalosporin**, **Bacitracin**, etc.

- 2. Alteration of cell membrane permeability:

Examples: **Nystatin**, **Polymyxin B**.

- 3. Inhibit protein synthesis:

- (i) 30S ribosome: **Tetracycline**, **Aminoglycosides** [AT]
- (ii) 50S ribosome: **Chloramphenicol**, **Macrolides** [CM]

- 4. Inhibit nucleic acid synthesis:

- (i) DNA synthesis: **Fluoroquinolones** (**Ciprofloxacin**), **Metronidazole**
- (ii) RNA synthesis: **Rifampicin**

3. Cell Wall Synthesis Inhibitor Drugs

(I) Beta-Lactam Drugs (contain 4-membered Lactam rings):

- 1. **Penicillin**
- 2. **Cephalosporin**
- 3. Carbapenem: **Imipenem**, **Meropenem**
- 4. Monobactam: **Aztreonam**

(II) Non-Beta-Lactam Drugs:

- 1. Glycopeptide antibiotics: **Vancomycin**, **Telavancin**, **Dalbavancin**
- 2. **Daptomycin**
- 3. **Bacitracin** (ত্বক কেটে গেলে বা ছিলে গেলে ব্যবহৃত হয় / used for cuts & abrasions)

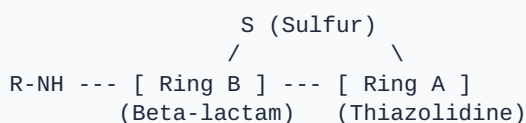
4. Penicillin: Sources & Chemical Structure

Sources of Penicillin:

- Obtained from fungus: *Penicillium notatum*
- Commercial source: *Penicillium chrysogenum*
- Natural penicillin: **Benzyl Penicillin** or **Penicillin G** (narrow spectrum and short acting)

Structure of Penicillin:

- (i) **Nucleus**: 6-aminopenicillanic acid (6-APA), which contains 2 rings.
- (ii) **Ring-A**: Contains Thiazolidine ring → responsible for hypersensitivity reactions.
- (iii) **Ring-B**: Called Beta-lactam ring → responsible for antibacterial action.



6-Aminopenicillanic Acid (6-APA) Core: Ring B (Beta-lactam) + Ring A (Thiazolidine with Sulfur)

- (iv) **Hydrolysis of beta-lactam ring**: Caused by beta-lactamase enzyme produced by some micro-organisms.

Beta-Lactamase Producing Organisms:

- (i) *Staphylococcus aureus* (*Staph. aureus*)
- (ii) *Escherichia coli* (*E. coli*)
- (iii) *Haemophilus influenzae*
- (iv) *Salmonella*
- (v) *Shigella*
- (vi) *Pseudomonas aeruginosa*
- (vii) *Streptococcus pyogenes*

Beta-Lactamase Inhibitor Drugs [OSPE]:

(Not antibiotics by themselves)

- (i) **Clavulanic acid**
- (ii) **Sulbactam**
- (iii) **Tazobactam**

5. Classification of Penicillin

A. On the Basis of Spectrum Coverage:

- **Narrow Spectrum:**

- **Benzyl Penicillin (Penicillin G)** — Acid labile
- **Phenoxymethyl Penicillin (Penicillin V)** — Acid stable

- **Anti-Staphylococcal Penicillin (Beta-Lactamase Resistant Penicillin):**

Examples: **Flucloxacillin, Cloxacillin, Nafcillin, Methicillin, Dicloxacillin**, etc.

Action: They work against *Staphylococcus aureus*.

- **Broad Spectrum:** Acts against both Gram (+ve) and Gram (-ve).

Examples: **Amoxicillin, Ampicillin, Talampicillin, Pivampicillin**.

- **Extended Spectrum (Anti-Pseudomonal Penicillin):**

Examples: **Carbenicillin, Ticarcillin, Azlocillin, Mezlocillin, Piperacillin**.

B. According to Source:

- (i) Natural: **Benzyl Penicillin**
- (ii) Semi-synthetic: **Phenoxymethyl Penicillin, Flucloxacillin, Amoxicillin, Ampicillin**

C. According to Duration of Action:

- **Short Acting:** **Penicillin G**, Broad Spectrum Penicillins.
- **Long Acting:** **Benzathine Penicillin**
 - Frequency: Monthly 1 time
 - Dose: 12 Lac IU (1.2 million IU)
 - Route: Deep IM (Gluteus maximus)

6. Indications of Penicillins

Benzyl Penicillin / Penicillin G:

- 1. Bacterial meningitis
- 2. Bacterial endocarditis
- 3. Syphilis
- 4. Gonorrhoea
- 5. Diphtheria
- 6. Gas gangrene

Phenoxymethyl Penicillin (Penicillin V):

- 1. Boil
- 2. Abscess
- 3. Pharyngitis
- 4. Tonsillitis
- 5. Prophylaxis against streptococcal infection following rheumatic fever

★ **Flucloxacillin (Anti-Staphylococcal):**

- 1. Skin and soft tissue infection — boil, carbuncle, abscess, cellulitis
- 2. Infected wound
- 3. Infected burn
- 4. Respiratory infection — pneumonia, sinusitis, lung abscess
- 5. UTI (Urinary Tract Infection)
- 6. Meningitis

Broad Spectrum (Amoxicillin / Ampicillin):

- 1. Respiratory Infection (RTI) — sinusitis, tonsillitis, otitis media, bronchitis, etc.
- 2. Enteric fever
- 3. Bacillary dysentery / Shigellosis
- 4. UTI

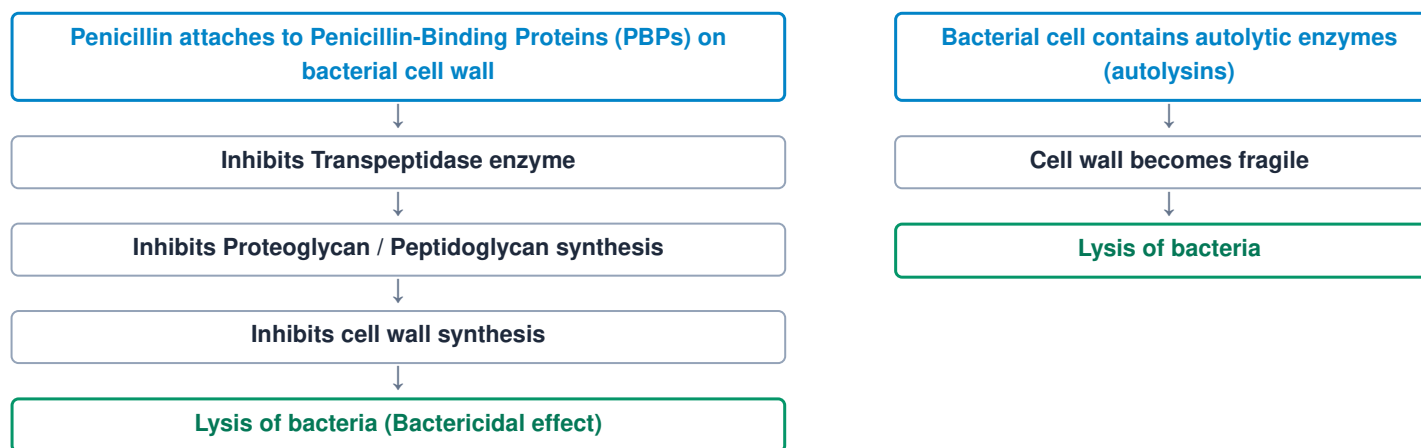
Long Acting / Benzathine Penicillin Deep IM Inj (Gluteus Maximus):

- 1. Treatment of Beta-hemolytic Streptococcal pharyngitis.
- 2. Prophylaxis of Rheumatic fever [Up to 5 years, treatment চলমান / continuous up to 18 years; Dose: 12 lac IU].
- 3. Treatment of Syphilis.

7. Mechanism of Action of Penicillin

Penicillin is a bactericidal antibiotic which inhibits bacterial cell wall synthesis by interfering with the transpeptidation reaction.

Penicillin inhibits the **3rd stage** of bacterial cell wall synthesis.



8. Difference between Amoxicillin and Ampicillin

Parameter	Amoxicillin	Ampicillin
Acid Stability	Acid stable	Acid stable
Oral Absorption	Better oral absorption	Incomplete oral absorption
Food Interference	No food interference	Food interference present (better on empty stomach)
Plasma Half-Life	1 hour	1 hour
Dosing Schedule	8 hourly / 3 times daily	6 hourly / 4 times daily

9. Adverse Effects of Penicillin

(I) Hypersensitivity Reaction (Type-I / IgE-Mediated Reaction):

- **1. Immediate:** Hypotension, bronchospasm, skin rash, fever.
- **2. Delayed:** Urticaria, fever, joint swelling (itch, raised skin), angioneurotic edema, hemolytic anemia.

(II) Other Adverse Effects:

- **GIT:** Diarrhoea, vomiting
- **Renal:** Acute interstitial nephritis
- **Superinfection**
- **Neurotoxicity**

10. Management of Hypersensitivity Reaction

Emergency Steps:

- 1. Stop drug therapy immediately
- 2. O₂ inhalation

- 3. Maintain IV fluid
- 4. Monitoring BP (Blood Pressure)
- 5. Specific Management:
 - (i) Inj. **Antihistamine** (IM/IV route) → to antagonize histamine.
 - (ii) Inj. **Hydrocortisone** (IV route) → to give permissive action.
 - (iii) Inj. **Adrenaline** (Subcutaneous route) → to cause bronchodilation & increase BP.

CRITICAL WARNING

Never give adrenaline IV or intracardiac; cause: patient can die immediately in this case.

11. Properties of Broad Spectrum Penicillin

- 1. Semi-synthetic
- 2. Acid resistant
- 3. Beta-lactamase sensitive
- 4. Short duration of action (6-8 hours)
- 5. Given with beta-lactamase inhibitor (e.g., **Clavulanic acid** / **Sulbactam** / **Tazobactam**)

Cephalosporins & Drug Combinations

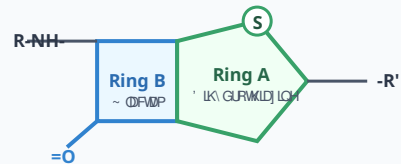
Structure, Generations, Spectrum & Clinical Rationales

1. Cephalosporin: Source & Structure

Source: Obtained from the fungus *Cephalosporium acremonium*. Now produced semi-synthetically to reduce adverse effects and broaden spectrum.

Core Structure: 7-aminocephalosporanic acid (7-ACA) nucleus consisting of two fused rings:

- **β-lactam Ring (Ring B):** Responsible for *antibacterial / bactericidal activity*.
- **Dihydrothiazine Ring (Ring A, with S):** Responsible for *pharmacokinetic properties* and chemical stability.



7-ACA Core: β-lactam + Dihydrothiazine ring

2. Advantages Over Penicillin

- **VIVA β-Lactamase Resistance:** Highly resistant to many bacterial β-lactamase enzymes compared to conventional penicillins.
- **Flexible Generational Spectrum:** Categorized into distinct generations with variable Gram (+ve)/(-ve) spectrum, pharmacokinetics, and Blood-Brain Barrier (BBB) penetration.
- **VIVA Hypersensitivity Profile:** Structurally related to penicillin (potential cross-allergenicity ~5-10%), but direct hypersensitivity is significantly lower than penicillin.

3. Classification & Generations of Cephalosporins

Key Rule: Gram (+ve) affinity decreases while Gram (-ve) spectrum and BBB permeability increase progressively from 1st to 4th generation.

1st Generation Cephalosporins

Pregnancy Safe Group

Spectrum: Excellent activity against Gram (+ve) cocci; modest activity against Gram (-ve).

BBB Penetration: **Does NOT cross BBB** — contraindicated/ineffective in meningitis.

Preparations:

- **Oral:** **Cephalexin**, **Cephradine**, **Cefadroxil**
- **Parenteral:** **Cefazolin**

✓ Clinical Indications

1. Uncomplicated upper and lower respiratory tract infections (URTI / LRTI)
2. Urinary Tract Infections (UTI — safe in pregnancy)
3. Cellulitis, soft tissue infections, and surgical wound prophylaxis

★ 2nd Generation

- **Spectrum:** Active against organisms covered by 1st generation drugs, but more active against Gram (-ve) and anaerobes.
- **BBB Penetration:** Only **Cefuroxime** can cross the Blood-Brain Barrier (BBB), so it is given in meningitis.
- **Preparations:**
 - Oral: **Cefaclor**, **Cefuroxime**
 - Parenteral: **Cefuroxime**, **Cefoxitin**

• Indications:

1. Upper and lower respiratory tract infection
2. UTI
3. Mixed anaerobic infection — peritonitis, pelvic infection (multiple organism infection)
4. Community-acquired pneumonia
5. Meningitis (only **Cefuroxime**)

★ 3rd Generation

Advantage of 3rd Generation

- **Spectrum:** Gram (+ve) coverage like 1st generation, but expanded Gram (-ve) activity more than 2nd generation.
- **BBB Penetration:** All cross the blood-brain barrier, so given in meningitis.
- **Preparations:**
 - Oral: **Cefixime**, **Ceftibuten**, **Cefpodoxime**
 - Parenteral: **Ceftriaxone**, **Cefoperazone**, **Ceftazidime**

• Indications:

1. Bacterial meningitis
2. Enteric fever
3. Septicemia
4. UTI, RTI
5. Gonorrhoea

★ 4th Generation

- **Features:** More resistant to β -lactamase enzyme; penetrates BBB.
- **Spectrum:** Active against Gram (+ve) and Gram (-ve) like 3rd generation, but also active against *Pseudomonas*, *Haemophilus*, and *Enterobacter*.
- **Preparation:** **Cefepime** (only parenteral preparation available).

• Indications: (Only in severe conditions)

1. Skin and soft tissue infection
2. Community-acquired pneumonia

4. Important Drug Combination Rationales

WRITTEN VIVA Rationale for the combination of Imipenem and Cilastatin :

- **Imipenem** is metabolized by the dehydropeptidase enzyme found in renal tubules. The inactive metabolite is potentially nephrotoxic.
- **Cilastatin** inhibits the dehydropeptidase enzyme.
- **Result:** Protects the parent drug (**Imipenem**), reduces nephrotoxicity, and prolongs the action of **Imipenem**.
- Therefore, in combination, **Imipenem** and **Cilastatin** increase efficacy and reduce toxicity.

WRITTEN VIVA Justify the combination of Amoxicillin and Clavulanic Acid :

- **Amoxicillin** is used against *Staphylococcus* which produces β -lactamase enzyme.
- β -lactamase enzyme destroys the β -lactam ring and inactivates **Amoxicillin**.
- **Clavulanic acid** inhibits this β -lactamase enzyme, keeping the antibacterial action of **Amoxicillin** intact.
- **Synergistic effect:** In combination, they achieve good tissue penetration and produce a powerful synergistic bactericidal effect.

Antimicrobial Classification, Principles, Resistance & Protein Synthesis Inhibition

1. Introduction to Antimicrobial Agents

Definition: The agents that are used against microbes are known as antimicrobial agents. They can kill or inhibit the growth/multiplication of micro-organisms.

Micro-organisms include: Bacteria, Virus, Fungus, Helminth, etc.

2. Classification of Antimicrobial Agents

A. According to the Type of Organism:

1. Antibacterial drugs
2. Antifungal drugs
3. Antiviral drugs
4. Antiprotozoal drugs
5. Anthelmintic drugs

Antibacterial Drugs:

1. **Penicillin**
2. **Cephalosporin**
3. **Ciprofloxacin**
4. **Metronidazole**
5. **Tetracycline**
6. **Streptomycin**
7. **Azithromycin**

Antifungal Drugs (Azoles / Conazoles & others):

1. **Fluconazole**
2. **Ketoconazole**
3. **Miconazole**
4. **Clotrimazole**
5. **Nystatin**
6. **Amphotericin B**
7. **Flucytosine**

Antiviral Drugs:

1. **Acyclovir**
2. **Ganciclovir**
3. **Lamivudine**
4. **Zidovudine**
5. **Interferon**

Anthelmintic Drugs:

1. **Albendazole**
2. **Mebendazole**
3. **Levamisole**

Antiprotozoal Drugs:

1. **Metronidazole**
2. **Diloxanide furoate**
3. **Dihydroemetine**
4. **Chloroquine**

3. Principles of Antimicrobial Therapy

1. **Make a proper diagnosis.**
2. **Remove the barrier to cure:** (e.g., incision and drainage of an abscess).
3. **Justification of therapy:** Determine whether antibiotics are indicated:
 - Acute infection → antibiotics are needed.
 - Chronic infection → antibiotics may or may not be needed.
4. **Selection of best drug:** According to patient's clinical condition, pharmacokinetic, and pharmacodynamic actions.
5. **Administration of drug in optimum dose, frequency, and appropriate route:** If dose frequency is not maintained, minimum inhibitory concentration (MIC) will not be sustained.
6. **Continuation of therapy:** Usually 5-7 days dose is prescribed; unnecessary long-term antibiotic use is dangerous.
7. **Advice to patient:** Antibiotics should generally be taken 1 hour before meals or 1-3 hours after meals (unless advised otherwise).
8. **Careful monitoring for side effects:**

SIDE EFFECTS / MONITORING

E.g., **Penicillin** and **Cephalosporin** causing GIT upset.

9. **Test for cure:**
 - **MIC:** Minimal Inhibitory Concentration.
 - **MBC:** Minimal Bactericidal Concentration.

4. Chemoprophylaxis

Definition: Prevention of disease and suppression of existing infections using antimicrobial agents.

INDICATIONS & EXAMPLES

1. **Prevention of primary infection:**
 - Rheumatic fever → Long-acting penicillin (**Benzathine Penicillin**).
 - Recurrent UTI → **Cotrimoxazole** (normally **Cephalosporin**).
2. **Prevention of opportunistic infection:**
 - Bacterial endocarditis after dental surgery → **Amoxicillin** or **Azithromycin** .
 - Prevent peritonitis after bowel surgery → **Cephalosporin** + **Metronidazole** .
3. **Suppression of existing infection:** E.g., Malaria, Tuberculosis (TB).

5. Antimicrobial Combinations

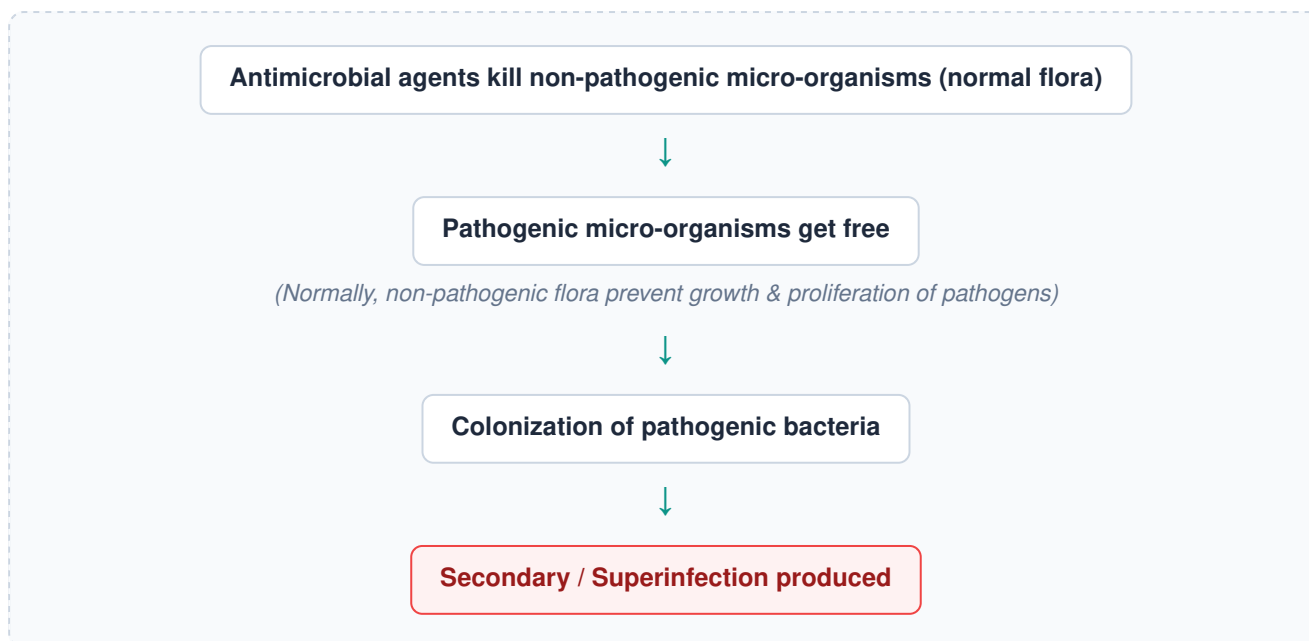
INDICATIONS / OBJECTIVES

1. **To avoid development of drug resistance:** E.g., Tuberculosis (TB) multi-drug regimen.
2. **To broaden the spectrum of coverage:** E.g., peritonitis after bowel surgery (**Cephalosporin** + **Metronidazole**).
3. **To obtain synergism / potentiation:** E.g., **Penicillin** + **Gentamicin** .
 - **Penicillin** covers Gram (+ve) organisms.
 - **Gentamicin** covers Gram (-ve) organisms.

6. Superinfection

Definition: Appearance of bacteriological and clinical evidence of a new infection during the treatment of a primary infection.

Mechanism of Superinfection:



Causative Drugs & Common Pathogens:

- **Offending Drugs:** **Amoxicillin** , **Ampicillin** , **Cephalosporins** , **Tetracycline** , etc.
- **Superinfecting Organisms:** *Candida albicans*, *Staphylococcus*, *Pseudomonas*, *Proteus*, etc.

7. Post-Antibiotic Effect (PAE) & Killing Dynamics

Definition: Persistent suppression of bacterial growth after limited exposure to an antimicrobial agent is called the post-antibiotic effect (PAE).

Types of Bacterial Killing:

1. Concentration-Dependent Killing:

- The rate and extent of killing increase with increasing drug concentration.
- **Dosing strategy:** A large dose is given once daily.
- **Examples:** Quinolones (**Ciprofloxacin**), Aminoglycosides (**Gentamicin**).

2. Time-Dependent Killing:

- The activity of the drug continues as long as serum concentrations remain greater than the MBC / MIC.
- **Dosing strategy:** Serum concentration should be maintained above the MIC for the entire inter-dosing interval.
- **Examples:** β -lactam drugs (**Penicillin** , **Cephalosporin**), Macrolides (**Azithromycin**).

8. Empiric Antimicrobial Therapy

Definition: Antimicrobial agents frequently used without knowing the specific pathogen responsible for the illness. Empiric therapy is initiated based on clinical experience and presumptive diagnosis.

Clinical Example: In upper respiratory tract infection (URTI), we empirically give **Cefixime** or respiratory fluoroquinolones (e.g., **Gemifloxacin** / **Levofloxacin**).

9. Opportunistic Infection

Definition: Infections that arise in immunocompromised patients caused by the body's normal microbial flora.

- **Predisposing Conditions:** AIDS patients, Leukemia, Pregnancy, Cancer chemotherapy, Cytotoxic drugs.
- **Common Organisms:**
 - *Candida* → Oral candidiasis, Vulvovaginitis.
 - *Staphylococcus*.
 - *Pneumocystis jirovecii* (formerly *P. carinii*).

10. Drug Resistance / Antimicrobial Resistance (AMR)

Definition: When inhibitory or killing effect of an antibiotic cannot be achieved without producing toxic effects on host cells, it is called bacterial resistance.

Types of Resistance:

1. Innate / Natural Resistance:

Some bacteria like coliforms and many other Gram (-ve) bacteria are naturally resistant to certain classes of antimicrobial agents (e.g., natural **Penicillin**).

2. Acquired Resistance:

• A. Genetic Resistance (Common):

1. **Chromosomal resistance:** Selective multiplication of resistant strains occurs via spontaneous mutation. Genetic transfer may occur through bacteriophages (virus-infected bacteria), particularly in *Staphylococcus*.

2. Extrachromosomal / Plasmid-mediated resistance:

- **(a) Enzymatic degradation of drug:** Micro-organisms produce enzymes that destroy active drug (e.g., *Staphylococcus* produces β -lactamase resistant to **penicillins**).
- **(b) Changes in permeability:** Micro-organism reduces permeability (e.g., **Tetracycline** fails to enter the bacterial cell).
- **(c) Alteration of target site:** Micro-organism alters drug target site, binding less effectively (e.g., **Aminoglycosides** cannot bind to 30S ribosomal target site).

• B. Non-Genetic Resistance (Less Common):

1. Developing an altered metabolic pathway.
2. Bacteria entering a resting / dormant state.
3. Surviving as protoplasts (loss of cell wall).
4. Active efflux pumping of the drug out of the cell.

11. Measures to Limit Drug Resistance

1. Avoid indiscriminate or inadequate use of drugs.
2. Restrict unnecessary use of antimicrobial combinations.
3. Prefer rapidly acting and selective drugs.
4. Limit the use of new drugs so that the effective lifespan of current drugs is prolonged.
5. Avoid drugs that promote the proliferation of resistant strains.

12. Protein Synthesis Inhibitor Antimicrobials

Sequence of Protein Synthesis in Bacterial Cells:

1. **mRNA Formation:** mRNA is formed on the DNA template in the nuclear region, carrying complementary nucleotide sequences from the DNA template.
2. **Initiation Complex:** After migrating from the nucleus to the cytoplasm, mRNA attaches to the 30S ribosomal subunit. The 50S subunit then binds to form the active 70S initiation complex to begin translation.
3. **tRNA Activation:** Specific amino acids in the cytoplasm are activated and attached to adaptor molecules of tRNA.
4. **Codon Recognition:** Aminoacyl-tRNA is transported to the Acceptor (A) site of the ribosome, guided by codon-anticodon pairing on mRNA.
5. **Peptide Bond Formation:** The nascent peptide chain attaches to the newly entered amino acid via peptide bond formation (catalyzed by peptidyl transferase).
6. **Translocation:** The elongated peptide chain shifts from the A site to the P (Peptidyl) site, and the ribosome advances along mRNA to expose the next codon for incoming aminoacyl-tRNA.
7. **Termination:** Finally, upon reaching a termination codon, translation stops, and the newly synthesized protein chain is released.

1. Tetracyclines: General Features & Chemistry

- **Broad Spectrum, bacteriostatic antibiotics.**
- **Chemical structure:** Contains 4 cyclohexane 6-membered rings fused together.
- **Spectrum:** Works against both aerobic and anaerobic Gram (+ve) and Gram (-ve) bacteria, including *Rickettsia*, *Chlamydia*, *Mycoplasma*, and some protozoa.
- **Mechanism:** Acts by inhibiting bacterial protein synthesis (binds to 30S ribosomal subunit).

Sources of Tetracyclines:

- (i) **Chlortetracycline** is obtained from *Streptomyces aureofaciens*.
- (ii) **Oxytetracycline** is obtained from *Streptomyces rimosus*.
- (iii) *These two are natural; the rest of all tetracyclines are semi-synthetic.*
- (iv) **Tetracycline** is produced from **Chlortetracycline** by catalytic dehalogenation.

2. Classification of Tetracyclines (According to Duration of Action)

1. Short Acting (6–8 hours, given 6 hourly):

Chlortetracycline, **Tetracycline**, **Oxytetracycline**.

2. Intermediate Acting (12 hours):

Demeclocycline, **Methacycline**.

3. Long Acting (16–18 hours):

Doxycycline, **Minocycline**, **Tigecycline** (36 hours, IV).

3. Pharmacokinetics of Tetracyclines

Absorption:

- Better absorbed in empty stomach *except* **Doxycycline**.
- **Site of absorption:** Upper intestine.
- **Chelation / Food Interaction:** Absorption is impaired due to concurrent administration of Ca^{2+} , Mg^{2+} , Fe^{2+} , Al^{3+} and dairy products (forms insoluble chelates with bivalent and trivalent cations).

Distribution:

- (i) Widely distributed in tissues and body fluids *except CSF*.
- (ii) Accumulates in the reticuloendothelial system (RES) cells of liver, spleen, and bone marrow.
- (iii) Deposits in bone, dentin, and enamel of un-erupted teeth → *contraindicated in pregnancy as it causes retardation of bony growth*.
- (iv) Crosses placenta and enters fetal circulation.

Excretion:

- (i) **Renal:** Mainly by glomerular filtration / secretion.
- (ii) **Bile:** **Doxycycline**, **Tigecycline**.
- (iii) **Breast milk.**

4. Mechanism of Action of Tetracyclines

Tetracycline enters bacterial cell partly by **passive diffusion** and partly by **energy-dependent active transport process**



Binds **reversibly** to the **30S ribosomal subunit**



Blocks the binding of **aminoacyl-tRNA** to the acceptor (A) site on mRNA-ribosomal complex



Prevents the addition of new amino acids to the growing peptide chain



Bacterial Protein Synthesis is Inhibited (Bacteriostatic Effect)

5. Indications of Tetracyclines

1. Diseases caused by Rickettsia, Chlamydia, Mycoplasma:

- *Rickettsial infections*: Q fever, Rickettsial fever, Typhus.
- *Mycoplasma infections*: Mycoplasma pneumonia.
- *Chlamydia infections*: Trachoma, Psittacosis, Pelvic Inflammatory Disease (PID), Lymphogranuloma venereum (LGV).

2. Respiratory tract infection.

3. Cholera (*Vibrio cholerae*) and other bacterial gastroenteritis.

4. Protozoal infections: Amoebic dysentery, Falciparum malaria.

5. Acne.

6. In combination regimen to treat *H. pylori* infection.

7. Plague.

6. Adverse Effects of Tetracyclines

1. **GIT upset:** Gastric irritation, abdominal discomfort, nausea, vomiting, diarrhoea.
2. **Teeth and Bones:** Tetracycline chelates calcium and forms *Tetracycline-Ca-orthophosphate complex* deposited into teeth and growing bone:
 - *In teeth:* Enamel dysplasia with pitting, brown discolouration, dental caries.
 - *In bones:* Growth retardation due to early fusion of epiphysis, bone deformity.
3. **Superinfection:** Sore throat, black hairy tongue by *Candida*, angular stomatitis.
4. **Hypersensitivity reaction:** Fever, rash, urticaria.
5. **Hepatotoxicity.**
6. **Nephrotoxicity.**
7. **Ototoxicity / Vestibular toxicity:** Caused by high-dose **Doxycycline** and **Minocycline**.
8. **Photosensitivity:** Especially with **Demeclocycline**.
9. **Nail pigmentation.**
10. **Fanconi syndrome.**

★ Fanconi Syndrome (Short Note):

This syndrome occurs when a patient ingests **outdated and degraded tetracycline**.

- **Features:** Nausea, vomiting, polyuria, polydipsia, proteinuria, glycosuria, acidosis.
- **Mechanism / Reason:** Toxic effect on proximal renal tubules → produces renal tubular acidosis (RTA).

Contraindications of Tetracyclines:

1. **Children below 8 years of age:** Due to growth retardation and tooth discolouration.
2. **Pregnancy:**
 - *Early pregnancy:* Teratogenicity.
 - *Late pregnancy:* Fetal bone deformity, tooth discolouration.
3. **Lactation.**
4. **Liver and renal disease.**
5. **Hypersensitivity to tetracycline.**

D/B: Tetracycline vs. Doxycycline:

Traits	Tetracycline	Doxycycline
Source	Semi-synthetic	Semi-synthetic
Potency	Low	High
Absorption	Moderate	High / Complete
Toxicity	More	Less
Dose / Half-life	Less / shorter (3 times a day)	More / longer (Once a day)

LECTURE 8 (L-8)

7. Aminoglycosides: Definition, Chemistry & Classification

Definition: Aminoglycosides are a group of antibiotics that contain a **hexose ring** to which **2 or more amino sugars** are attached by **glycosidic bonds**.

They act mainly against **Gram (-ve) aerobic micro-organisms**, and are also active against *Pseudomonas* and *Mycobacterium tuberculosis*.

Preparations: They have no effective oral absorption (only parenteral and topical preparations are present).

Exception: **Neomycin** is given orally for bowel sterilization before surgery (অপারেশনের আগে বাওয়েল ক্লিয়ার করার জন্য).

According to Route of Administration:

• (I) Systemic:

1. Streptomycin
2. Kanamycin
3. Tobramycin
4. Paromomycin
5. Gentamicin
6. Amikacin

• (II) Topical MCQ :

1. Neomycin (Soframycin / SAB)
2. Framycetin

According to Source:

• 1. From *Streptomyces*:

- (i) Streptomycin
- (ii) Neomycin
- (iii) Tobramycin
- (iv) Kanamycin

• 2. From *Micromonospora*:

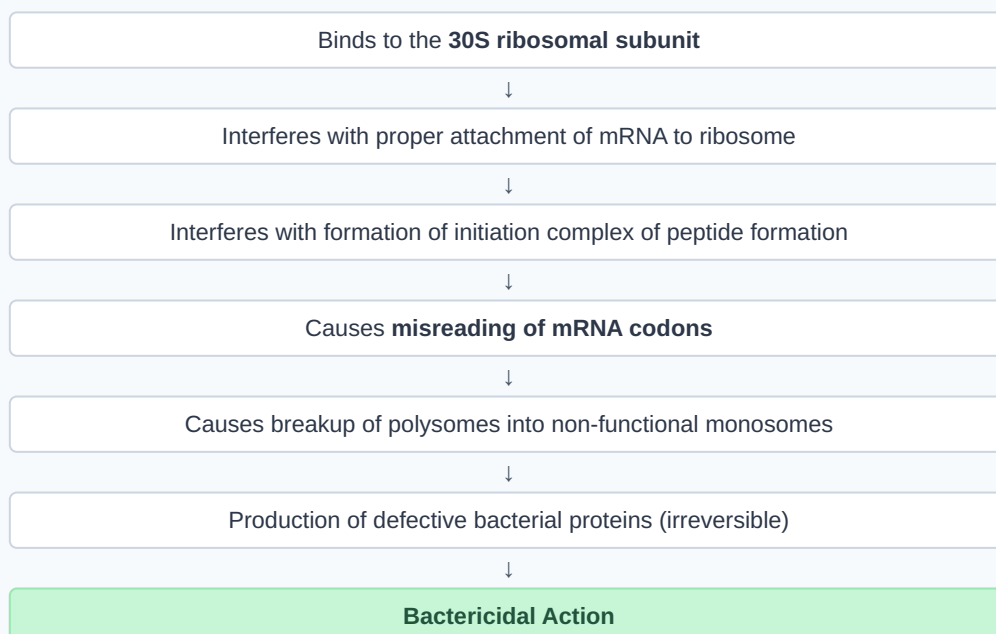
- (i) Gentamicin
- (ii) Amikacin
- (iii) Netilmicin

8. Common Properties of Aminoglycosides

1. Contain a hexose ring attached to 2 or more amino sugars by glycosidic bonds.
2. Water soluble, more active in **alkaline pH**.
3. Oral absorption is poor; distribution is mainly extracellular.
4. Highly polar, polycationic → not absorbed in gut; oral administration does not produce systemic action.
5. Excreted through kidney mainly by **glomerular filtration**.
6. They are **bactericidal** and act by **irreversible inhibition of protein synthesis**.
7. Highly effective against Gram (-ve) aerobic bacteria.
8. **Cannot cross Blood-Brain Barrier (BBB)**, but can cross placental barrier (teratogenic).
9. They all show common toxicities: **ototoxicity, nephrotoxicity, and neurotoxicity**.
10. They have a **low therapeutic index**.
11. Resistance develops quickly and shows cross-resistance among different aminoglycosides.

9. Mechanism of Action of Aminoglycosides

Aminoglycosides are **bactericidal** antibiotics acting by **irreversible inhibition of bacterial protein synthesis**.



10. Indications of Specific Aminoglycosides

★ **Gentamicin** :

1. Septicemia (blood sepsis)
2. Bacteremia
3. Abdominal & pelvic sepsis
4. Subacute bacterial endocarditis (SBE)
5. Peritonitis
6. Infected burns
7. Osteomyelitis
8. Severe UTI
9. Severe pneumonia

★ **Streptomycin** :

1. Tuberculosis (TB)
2. Plague
3. Tularemia
4. Brucellosis

★ **Neomycin** :

1. Infected burns, wounds, ulcers (topical)
2. Preparation / sterilization of bowel before surgery
3. Hepatic encephalopathy / coma
4. External ear infection (drops, ointment)
5. Conjunctivitis

11. Adverse Effects & Toxicities of Aminoglycosides

Ototoxic Aminoglycosides:

- **Streptomycin** , **Neomycin** (একসাথে দেওয়া যাবে না / Do not combine)
- **Gentamicin**
- **Amikacin**

Nephrotoxic Aminoglycosides:

- **Gentamicin**
- **Neomycin**
- **Tobramycin**

Detailed Toxicities:

1. **Ototoxicity:** Drug enters perilymph and endolymph; no back-diffusion into blood occurs → high drug concentration in labyrinthine fluid → damages the **8th cranial nerve**:
 - *Auditory / Cochlear damage (Reversible):* Causes tinnitus and hearing loss (caused by high-concentration **Amikacin** , **Kanamycin**).
 - *Vestibular damage (Irreversible):* Causes vertigo, ataxia, loss of balance (caused by **Gentamicin** , **Streptomycin**).
2. **Nephrotoxicity:** Attains high concentrations in renal cortex leading to tubular necrosis (caused by **Neomycin** , **Tobramycin** , **Gentamicin**).
3. **Neuromuscular blockade:** In high doses, causes respiratory muscle paralysis.
4. **Hypersensitivity reaction.**
5. ★ **Teratogenesis:** Causes 8th cranial nerve damage leading to deafness in newborns.

Contraindications:

1. Pregnancy
2. Renal impairment
3. Hypersensitive patients
4. Myasthenia gravis (muscle weakness problem)

Limitations & Precautions:

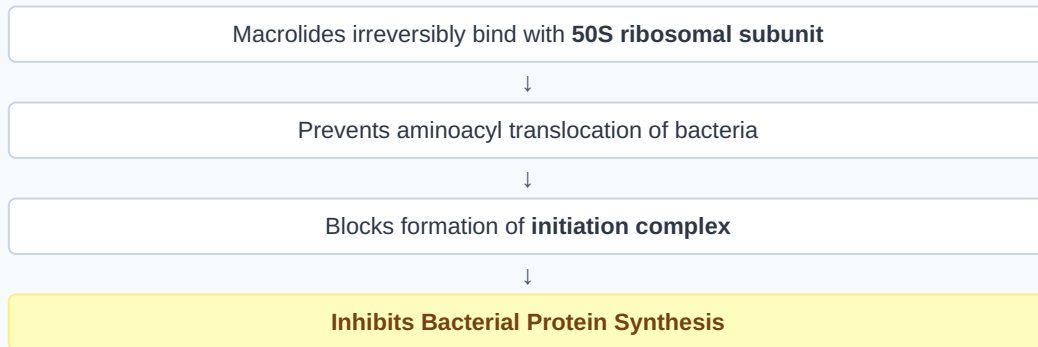
1. Low therapeutic index.
2. High toxicity in elderly patients and patients with renal insufficiency.
3. Careful use with other concomitant nephrotoxic / ototoxic drugs like NSAIDs, **Amphotericin B** .
4. Cannot be used in pregnant women.

1. Macrolides: General Features & Mechanism

Definition: Macrolides are a group of antibiotics having a **macrocyclic lactone ring** (usually containing 14–16 carbon lactone ring) to which one or more **deoxy sugars** are attached.

- **Mechanism:** They act by inhibiting bacterial protein synthesis by binding to the **50S ribosomal subunit**.
- All protein synthesis inhibitors are **bacteriostatic** except aminoglycosides.
- (At low doses they are bacteriostatic; at high doses some of them become **bactericidal** in nature.)
- **Examples:** **Erythromycin**, **Azithromycin**, **Clarithromycin**, **Roxithromycin**.

Mechanism of Action:



2. Erythromycin

- **Source:** Obtained from *Streptomyces erythreus*.
- **Spectrum:** Narrow spectrum antibiotic. Usually bacteriostatic at low doses, but at high doses it is bactericidal in nature.
- **Antimicrobial Activity:** Effective against Gram (+ve) micro-organisms, specially *Streptococci*, *Staphylococci*, *Pneumococci*, etc. Also acts against Gram (–ve) micro-organisms like *Neisseria*, *Campylobacter*, etc.
- **Nature & Formulation:** It is acid labile (coated, so that it doesn't get destroyed by gastric juice; thus achieving better absorption at site of action). Administered with enteric coating or with acid-resistant substances like stearate, estolate, etc. for better absorption.

Indications of Erythromycin:

1. ★ Pharyngitis, tonsillitis, mastoiditis
2. ★ Diphtheria (drug of choice)
3. Atypical pneumonia caused by *Mycoplasma*
4. Whooping cough
5. ★ *Campylobacter* enteritis
6. Used as a penicillin substitute in penicillin-allergic patients
7. Acne

Adverse Effects of Erythromycin:

1. **Hypersensitivity reaction:** Fever, rash, urticaria, eosinophilia
2. **Liver:** Cholestatic jaundice
3. **GIT:** Epigastric pain, anorexia, nausea, vomiting, diarrhoea
4. **Drug interactions:** **Erythromycin** inhibits Cytochrome P450 enzyme → increases serum concentration & prolongs action of **Warfarin**, **Theophylline**, **Methylprednisolone**, etc.

3. Azithromycin & Clarithromycin

★ Azithromycin :

It is a commonly used and good antibiotic.

Indications:

1. **Respiratory Tract Infection (RTI):** Pharyngitis, tonsillitis, otitis media, sinusitis, acute and chronic bronchitis, etc.
2. Skin and soft tissue infection
3. Enteric fever
4. Cervicitis, Uveitis
5. Urethritis
6. Acne (**Erythromycin** , **Tetracycline** , **Azithromycin**)

Adverse Effects & Features:

- **GIT:** Epigastric pain, anorexia, nausea, vomiting, diarrhoea (severity is less than erythromycin).
- **Dosing:** Given once daily; tissue concentration is high, so released slowly.
Ex: Acute watery diarrhoea → 1st day (2 tablets = 1g), 2nd and 3rd day (1 tab daily; net 4 tablets, max 5 days).
- **Half-life:** 2–4 days.
- **Administration:** 1 hour before meal or 2 hours after meal.
- **No significant drug interaction.**

★ Clarithromycin :

Indications:

1. **Respiratory Tract Infection:** Sinusitis, tonsillitis, bronchitis, otitis media, etc.
2. **Community-acquired pneumonia:** (affected from large crowded area; not single organism).
3. **PUD (Peptic Ulcer Disease) Triple Therapy:** 1 gm twice daily for 2 weeks.
4. **Tularemia.**

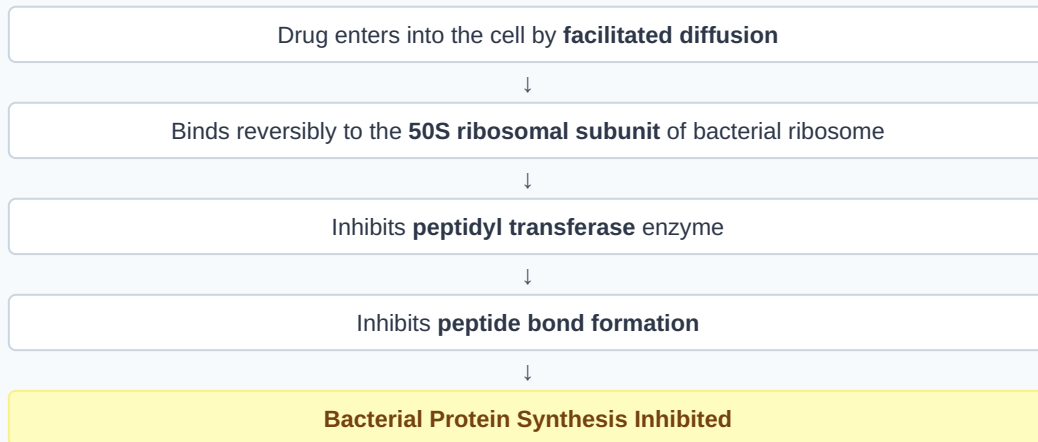
4. Comparison: Erythromycin vs. Azithromycin vs. Clarithromycin

Traits	Erythromycin	Azithromycin	Clarithromycin
Structure	Contains 14-lactone ring	15-lactone ring (Azalide)	14-lactone ring
Half-life	1.5 hours	2 – 4 days	6 hours
Dose & Duration	(250–500) mg 6 hourly, (5–7) days	500 mg once daily, (3–5) days	(250–500) mg 12 hourly (Acne: 2 weeks – even 1 month)
Cost	Cheap	Costly	Costly
Excretion	Biliary excretion more	Bile and Feces	Biliary excretion more
Mode of Action	Bacteriostatic (low dose) Bactericidal (high dose)	Bacteriostatic	Bacteriostatic (low dose) Bactericidal (high dose)

5. Chloramphenicol

- **Source:** *Streptomyces venezuelae*.
- **Broad spectrum and bacteriostatic antibiotic.**
- **Spectrum:** Active against Gram (+ve) and Gram (–ve) organisms, and also against *Rickettsia*.

Mechanism of Action:



Indications:

Use of **Chloramphenicol** is now limited because of its potential toxicities and presence of newer and safer antibiotics.

- **Topical use:** Used as ointment / drops in **bacterial conjunctivitis** and **ear infection**.

Adverse Effects:

1. **GIT effect:** Nausea, vomiting, diarrhoea.
2. **Bone marrow suppression:**
 - *Dose-related toxicities (high dose):* Anemia, leukopenia, thrombocytopenia.
 - *Dose-unrelated (idiosyncratic reaction):* **Aplastic anemia**.
3. **Toxicity for newborn:** In premature babies, causes "**Gray Baby Syndrome**" at high doses.
4. **Hypersensitivity reaction.**
5. **Drug interactions:** Inhibits Cytochrome P450 enzyme → inhibits metabolism → increases serum concentration → prolonged action & toxic effects of **Warfarin**, **Phenytoin**, **Tolbutamide**, etc.

★ **Gray Baby Syndrome (Short Note / Viva):**

In premature babies, high doses of chloramphenicol cause Gray Baby Syndrome. It usually begins within 2–9 days with vomiting, flaccidity, hypothermia, gray color skin/cyanosis, shock, cardiovascular collapse, and ultimately death. Patient may die on the 2nd day.

Causes / Mechanism:

- (i) **Inadequate metabolism** of chloramphenicol by *glucuronyl transferase* (premature hepatic enzymes are insufficient to metabolize drugs).
- (ii) **Inadequate excretion** of unconjugated drug due to undeveloped renal function in newborns.

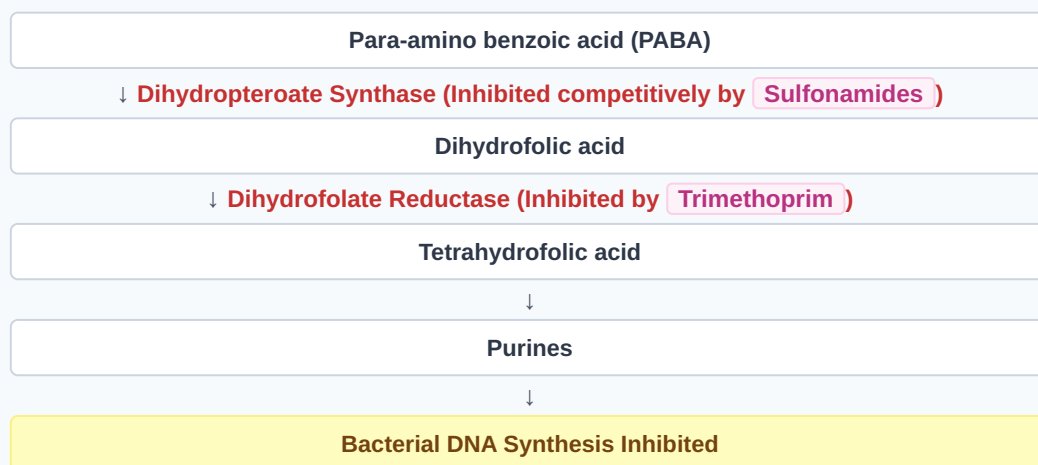
6. Sulfonamides: General Features & Classification

- Broad spectrum bacteriostatic antibiotic.
- It is the first chemotherapeutic agent, derived from **Prontosil**.
- Why use is limited nowadays:
 - (i) More adverse effects.
 - (ii) Rapidly developing bacterial resistance.
 - (iii) Presence of more safe and newer antibiotics.
- Still, the combination of **sulfonamide + trimethoprim** is widely used against a variety of infections (indirectly inhibits DNA synthesis).

Classification of Sulfonamides:

- (I) Systemic (used in oral / injection):
 - 1. Short Acting (4–8 hours): **Sulfadiazine**
 - 2. Intermediate Acting (8–12 hours): **Sulfamethoxazole**, **Sulfasalazine**
 - 3. Long Acting (7 days): **Sulfadoxine**
- (II) Topical:
 - **Sulfacetamide** (eye drop)
 - **Silver Sulfadiazine** (used as ointment/drop in infected burns, leg ulcers)

7. Mechanism of Action of Sulfonamides



⇒ Sulfonamides indirectly inhibit bacterial DNA synthesis by inhibiting the metabolic pathway of folate synthesis.

8. Indications & Adverse Effects of Sulfonamides

Indications:

1. UTI
2. Meningococcal meningitis (*Streptococcus meningitidis* / *Neisseria meningitidis*)
3. Chloroquine-resistant malaria
4. Ulcerative colitis
5. Inflammatory bowel disease
6. Infected burns, leg ulcers (**Silver Sulfadiazine**)
7. Conjunctivitis (**Sulfacetamide**)
8. Lymphogranuloma venereum (LGV)

Adverse Effects:

1. **Renal toxicities:** Sulfonamide precipitates in urine → produces crystalluria, hematuria, renal colic, renal tubular acidosis.
2. **Hypersensitivity reaction:** Rash, fever, urticaria, **Stevens-Johnson syndrome** (sloughing out of skin / fatal condition).
3. **GIT:** Anorexia, nausea, abdominal cramp, vomiting.
4. **Hemolytic anemia, aplastic anemia.**
5. **Teratogenesis:** Produces **kernicterus** in newborns and fetus (displaces bilirubin from albumin).

1. Cotrimoxazole: Composition & Advantages

Definition & Composition: **Cotrimoxazole** is a combination of **Sulfamethoxazole** + **Trimethoprim**.

Half-life: 10 hours.

Advantages:

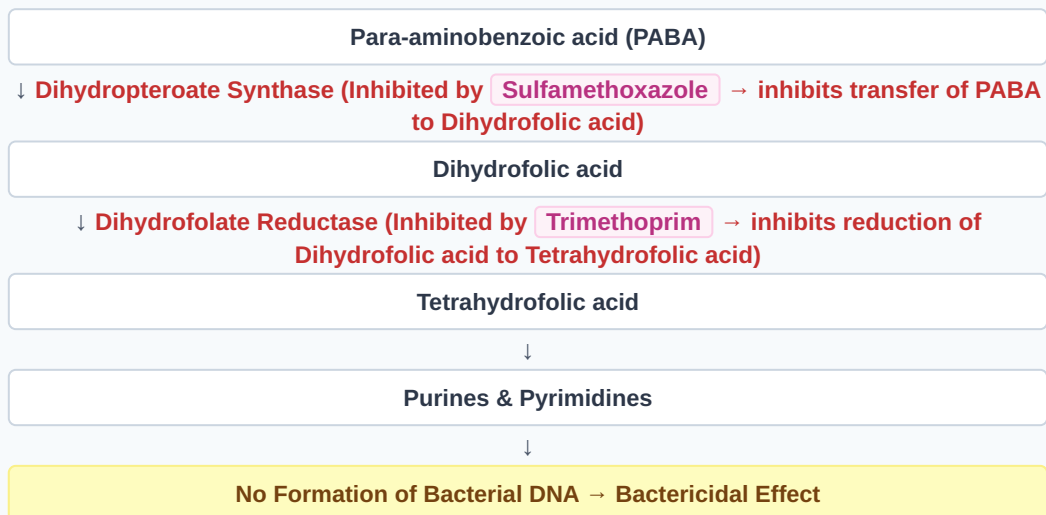
1. To obtain synergistic effects
2. **Bactericidal action** (individual drugs are bacteriostatic, but in combination they become bactericidal)
3. **Broad spectrum**

Antimicrobial Activity:

Active against *Streptococcus*, *Staphylococcus*, *Corynebacterium*, *Escherichia coli*, *Salmonella*, *Shigella*, etc.

2. Mechanism of Action of Cotrimoxazole

The combination of **Sulfamethoxazole** and **Trimethoprim** produces **synergistic effects by sequential block** in the bacterial metabolic pathway of folic acid synthesis.



3. Indications of Cotrimoxazole

1. **Respiratory tract infection:** Sinusitis, otitis media, pharyngitis, pneumonia, etc.
2. **UTI (Urinary Tract Infection)**
3. **Enteric fever**
4. **Infected burns**
5. **GIT:** Diarrhoea, dysentery
6. **Skin and soft-tissue infection**

Adverse Effects:

1. **Renal toxicities:** Crystalluria, hematuria, renal tubular acidosis.
2. **Hypersensitivity reaction:** Rash, fever, **Stevens-Johnson syndrome**, toxic epidermal necrolysis (TEN).
3. **Blood:** Hemolytic anemia, pancytopenia.
4. **GIT:** Anorexia, nausea, vomiting, diarrhoea.

Contraindications:

1. **Pregnancy**
2. **Lactation**
3. **Severe renal disease**

4. Important Written & Viva Question

WRITTEN / VIVA Justify / Rationale for the combination of Sulfonamide and Trimethoprim:

- **Cotrimoxazole** is a fixed-dose combination of **Trimethoprim** (80 mg) and **Sulfamethoxazole** (400 mg).
- The ratio of these drugs is **1 : 5**.
- Individual drugs are **bacteriostatic**, but in combination they are **bactericidal**.
- **Reasons for combining these two drugs:**
 1. Both have nearly the **same half-life (10 hours)**.
 2. They act on the **same metabolic pathway** (sequential blockade of folic acid synthesis).
- **Why the ratio is disproportionate (1 : 5):**

Trimethoprim is more lipid-soluble and has a larger volume of distribution. For this reason, **1 part of Trimethoprim** is given with **5 parts of Sulfamethoxazole** to achieve optimal synergistic plasma concentrations (1 : 20 ratio in blood).

1. General Properties & Infection Sites

Properties of *Mycobacterium tuberculosis*:

- Aerobic organism
- Residing both intracellularly and extracellularly
- Can remain in a dormant stage for a long time
- Drugs penetrate these cells poorly
- Rapidly develops resistance, so treatment response is very slow

⇒ Therefore, drugs must be taken regularly and therapy continued for a long time (e.g., DOTS center under government supervision to avoid default and resistance).

Cidal vs. Static Action:

All Anti-TB drugs are **tuberculocidal** except

Ethambutol (which is **tuberculostatic**).

MCQ / VIVA Sites of Infection:

- **1. Pulmonary:** Lung
- **2. Extra-pulmonary:** Bone, Kidney, Skin, Liver, GI tract

2. Classification of Anti-TB Drugs

1st Line Drugs — FDC (Fixed Dose Combination):

1. **Rifampicin** — **TUBERCULOCIDAL**
2. **Isoniazid** (**INH**) — **TUBERCULOCIDAL**
3. **Ethambutol** — **TUBERCULOSTATIC**
4. **Pyrazinamide** — **TUBERCULOCIDAL**
5. **Streptomycin** — **TUBERCULOCIDAL** (Mainly IM / IV)

2nd Line Drugs:

1. **Thiacetazone**
2. **Para-aminosalicylic acid** (**PAS**)
3. **Kanamycin**
4. **Amikacin**
5. **Capreomycin**
6. **Cycloserine**

Newer Drugs (2nd Line Group):

1. **Ciprofloxacin**
2. **Levofloxacin**
3. **Clarithromycin**
4. **Rifabutin**

3. Principles of Combined Anti-Tubercular Therapy (ATT)

1. Broaden the spectrum / Kill wide range of actively multiplying bacilli:

- Kill large range of actively multiplying bacilli by **Isoniazid** (**INH**) → reduces bacterial load rapidly.
- Treat semi-dormant bacilli by **Rifampicin** and **Pyrazinamide**.

2. To prevent the emergence of drug resistance: Achieved by using **Isoniazid** and **Rifampicin** together.

3. Shorten the duration of treatment: Achieved by adding **Pyrazinamide** (enables 6-month short-course regimen).

4. First-Line Drug Profiles & Mechanisms

1. Isoniazid (INH) Enzyme Inhibitor

It is a tuberculocidal prodrug active against *Mycobacterium tuberculosis*, activated by the mycobacterial **catalase-peroxidase** enzyme.

Mechanism of Action: Inhibits the synthesis of **mycolic acid** (an essential component of the mycobacterial cell wall) → kills the bacteria.

Adverse Effects:

1. Peripheral Neuropathy:

- Occurs in **Slow Acetylators** due to relative deficiency of pyridoxine.
- **INH** is structurally similar to Vitamin B6 (**Pyridoxine**), promoting excretion of pyridoxine.
- *Reversed / Prevented by:* Administration of **Pyridoxine 10 mg/day**.

2. Hepatitis:

- Occurs more frequently in **Rapid Acetylators**.
- **INH** is metabolized in the liver by acetylation and hydrolysis into hepatotoxic metabolites.

3. **CNS Effects:** Psychosis, memory loss, convulsion.

4. **Drug Interaction:** Inhibits hepatic microsomal enzymes.

2. Rifampicin Enzyme Inducer

It is a tuberculocidal agent.

Mechanism of Action: Inhibits bacterial RNA synthesis by binding with **DNA-dependent RNA polymerase** enzyme → prevents bacterial protein synthesis → bactericidal action.

Indications:

1. Tuberculosis
2. Leprosy
3. Prophylaxis of meningococcal and *H. influenzae* meningitis
4. Streptococcal endocarditis
5. MRSA infection

Adverse Effects:

1. **Flu-like syndrome:** Fever, chills, myalgia
2. **Hepatitis, Cholestatic jaundice** (Hepatotoxic)
3. **Harmless orange/red colour** of body fluids (urine, sweat, tears)
4. Thrombocytopenia
5. Nephritis
6. **Hepatic enzyme inducer:** Decreases levels of oral contraceptives, warfarin, etc.

3. Ethambutol TUBERCULOSTATIC

Mechanism of Action: Inhibits mycobacterial **arabinosyl transferase** enzyme → blocks the polymerization reaction of arabinoglycan (an essential cell wall component) → cell wall synthesis is inhibited.

Adverse Effects (Eye Problems & Joint):

1. Optic neuritis
2. Red-green colour blindness
3. Arthralgia / Hyperuricemia

Hepatotoxic Anti-TB Drugs: **Isoniazid**, **Rifampicin**, **Pyrazinamide**.
(Serum ALT / AST monitoring is mandatory for patients on these drugs.)

5. Anti-TB Treatment Regimens (Categories)

Category	Patient Types	Duration	Drug Regimen
Category I	New Case, Smear (+ve), Pulmonary TB, Extra-pulmonary TB	6 Months	• 2 months (Intensive): 4 FDC (R , I , P , E) • 4 months (Continuation): 3 FDC (R , I , E)
Category II	Relapsing case, Failure case, Smear (+ve) treatment failure	8 Months	• 2 months: 5 FDC (R , I , P , E + Streptomycin) • 1 month: 4 FDC (R , I , P , E) • 5 months: 3 FDC (R , I , E) (Pyridoxine added for 8 months)
Category III	Smear negative (-ve), Extra-pulmonary TB	6 Months	• 2 months: 4 FDC (R , I , P , E) • 4 months: 3 FDC (R , I , E)

6. Special Topics (DOTS, MDR-TB & XDR-TB)

★ DOTS (Directly Observed Treatment, Short-Course):

- Internationally recommended strategy for TB control.
- Most cost-effective way to stop the spread of TB in high-incidence communities by ensuring complete cure and preventing resistance.

★ MDR-TB (Multi-Drug Resistant Tuberculosis):

Defined as resistance to **both Isoniazid (INH)** and **Rifampicin** (and may include resistance to any other 1st line drugs).

- In case of INH resistance alone: **Rifampicin** + **Ethambutol** + **Pyrazinamide** given for 12 months (with **Pyridoxine** supplementation).

• In case of combined INH & Rifampicin resistance (2nd Line Regimen):

1. **Pyrazinamide** / **Ethambutol**
2. Aminoglycosides: **Streptomycin** / **Kanamycin** / **Amikacin**
3. **Ciprofloxacin** and/or **Ethionamide**

★ XDR-TB (Extensively Drug-Resistant Tuberculosis):

Rare form of MDR-TB that is resistant to **Isoniazid** + **Rifampicin**, PLUS any **Fluoroquinolone**, AND at least one of the three injectable second-line drugs (**Amikacin** / **Kanamycin** / **Capreomycin**).

7. Role of Corticosteroids in Tuberculosis

Corticosteroids are used as an **adjunct** in severe forms of TB to prevent inflammatory complications.

Prevention of Complications:

- Constrictive pericarditis
- Hydrocephalus
- Focal neurological deficits
- Pleural adhesions
- Intestinal strictures

Indications of Steroids in TB:

• Absolute Indication:

- Bilateral adrenal TB

• Relative Indications:

1. Miliary TB / severe pulmonary TB (millet-seed appearance on X-ray)
2. Severe hypersensitivity reactions to anti-TB drugs
3. Meningeal TB, Renal TB
4. TB with pleural effusion
5. TB with AIDS

⚠ CONTRAINDICATION:

Intestinal TB — strictly contraindicated due to the high risk of *silent intestinal perforation*.



1. Introduction to Hematinics & Iron Distribution

Definition: Hematinics are the drugs which help in the qualitative and quantitative production and maturation of RBC (blood content / RBC maturation / production).

Common Hematinics are:

1. **Iron**
2. **Vitamin B12** (Cyanocobalamin)
3. **Folic acid**

Iron Physiology & Body Distribution:

- Iron is an essential mineral that carries O_2 and forms part of Hemoglobin (Hb) in our RBCs and myoglobin in muscle.
- **Total iron in our body:** 3 – 5 gm.
- **Hb contains:** About 2/3rd of total body Fe.
- **Stores comprise:** About 1/3rd as *ferritin* and *hemosiderin* in liver, bone marrow, spleen, and muscle.
- Iron is stored in liver, spleen, and bone marrow in the form of **ferritin**.
- **5 – 10%** iron is present in tissues throughout the body in **myoglobin form**.
- Normal adult human absorbs **5 – 10%** of dietary Fe.

Sources of Iron:

- **Animal Source:** Liver, egg yolk, red meat, fish, etc.
- **Vegetable Source:** Green leafy vegetables, lentils, soybean, beans, raisin, green bananas, cereals, pears.

Daily Requirements:

- **Children:** 10 – 15 mg
- **Adult male:** 5 – 10 mg
- **Adult female:** 15 – 18 mg (*more due to menstruation/ pregnancy*)

2. Iron Preparations & Indications of Oral Iron

Oral Iron Preparations:

1. **Ferrous sulphate** (tab)
2. **Ferrous gluconate** (tab)
3. **Ferrous fumarate** (tab)
4. **Ferrous succinate** (tab)

Parenteral Iron Preparations:

1. **Iron dextran** (IV)
2. **Iron sucrose** (IV)
3. **Iron sorbitol** (IM)

Indications of Oral Iron:

1. Chronic Blood Loss:

- 1. GI bleeding — due to hookworm infestations, bleeding peptic ulcer, haemorrhoids
- 2. Carcinoma (various types of cancer)
- 3. Menorrhagia

2. Increased Physiological Demand:

- 1. Pregnancy, lactating mother
- 2. Child during growing period
- 3. Patients with chronic kidney disease (CKD)

3. Inadequate Iron Absorption:

- 1. Coeliac disease
- 2. Post-gastrectomy
- 3. Malabsorption syndrome

4. Less dietary intake

Treatment Strategy & Adverse Effects of Oral Iron:

- **Dose:** (200 – 400) mg oral iron should be given daily to correct iron deficiency anemia.
- **Duration:** Oral treatment should be continued for **3 – 6 months** (correction of anemia + replenish iron stores in the last 3 months).

Adverse Effects of Oral Iron:

1. Nausea, vomiting, epigastric discomfort
2. Abdominal cramp
3. Diarrhoea / Constipation
4. **Black colour stool** (due to unabsorbed iron forming iron sulfide)

3. Parenteral Iron: Indications & Adverse Effects

Indications of Parenteral Iron (Emergency Condition):

1. Patient with excessive chronic blood loss
2. Patient with post-gastrectomy state
3. Inflammatory bowel disease (IBD)
4. Malabsorption syndrome
5. Anemia not corrected with oral therapy / severe intolerance to oral iron

Adverse Effects of Parenteral Iron:

• (i) Local:

- Pain at the site of IM injection
- Abscess formation
- Skin staining / pigmentation

• (ii) Systemic:

- Fever, headache, arthralgia
- Flushing, palpitation, dyspnea
- Thromboembolism / Thrombophlebitis
- ★ **Anaphylactoid reaction** → leading to vascular collapse and death.

4. Iron Toxicity / Overdose / Poisoning

Acute Toxicity / Poisoning:

Signs & Symptoms:

1. Severe abdominal pain
2. Gastroenteritis and vomiting
3. Bloody diarrhoea followed by shock, dyspnea
4. Severe metabolic acidosis
5. Coma, death

Treatment of Iron Poisoning:

1. **Immediate Treatment:** Iron should be chelated by **Desferrioxamine** (in blood, stomach, and intestine).
2. **Gastric Decontamination:** Induced vomiting / gastric lavage with raw egg and milk (in acute poisoning).

★ Desferrioxamine (Specific Antidote):

- It is the **specific antidote** for iron poisoning.
- **Dose:** (1 – 2) gm IM. But in severe cases, it may be given by IV route / infusion.

Chronic Poisoning:

Hemochromatosis:

- Chronic toxicity is known as **hemochromatosis / hemosiderosis** where iron is deposited into heart, liver, pancreas, and other organs.
- **Causes:** Occurs due to repeated blood transfusions (e.g., in Thalassemia patients) or genetic defect.

5. Iron Absorption & Transport Mechanism

Site: Duodenum and upper jejunum (predominantly).

Iron is absorbed in the GIT in the form of **Fe²⁺ (ferrous form)**. So, all ferric (Fe³⁺) form must be converted into Fe²⁺ form for absorption in GIT.

Iron Absorption is a Two-Stage Process:

1. **Active Process:** Rapid absorption of Fe takes place at the brush border of the intestinal epithelium by active transport.
2. **In Mucosal Border:** Fe²⁺ is oxidized to Fe³⁺. Then Fe³⁺ binds with transferrin. Transferrin-bound Fe is transferred from cell to plasma or stored as **ferritin** in the mucosal cell.

Iron Transport: In plasma, iron is transported in the form of **transferrin**.

Factors Increasing Fe Absorption:

1. Low body stores of Fe
2. Condition associated with increased rate of erythropoiesis
3. Ascorbic acid (Vitamin C → reduces Fe³⁺ to Fe²⁺)
4. Intake of inorganic Fe
5. Pathological conditions — hemochromatosis, chronic liver disease, pancreatic insufficiency, etc.
6. Fasting condition
7. Alcohol intake

Factors Decreasing Fe Absorption:

1. Malabsorbing syndrome
2. Diarrhoeal disease
3. Subtotal gastrectomy
4. Surgical removal of upper intestine
5. Antacid intake, **Tetracycline** intake (contain bivalent/trivalent molecules that chelate iron)
6. Food intake along with Fe (phytates, phosphates, tea/tannins)
7. Chronic infection
8. High body stores of Fe / Fe overload

6. Vitamin B12 (Cyanocobalamin)

Physiology: **Vitamin B12** and **Folic acid** are required for the synthesis of DNA. Deficiency causes abnormal maturation and production of RBC leading to **megaloblastic anemia**.

Sources:

- **Rich Source:** Liver, kidney, spleen, heart.
- **Less in:** Egg yolk, fish, meat, milk.

Requirement & Absorption:

- **Daily Requirement:** (5 – 30) microgram daily.
- **Absorption Site: Terminal ileum** (requires intrinsic factor of Castle).

Indications of Vitamin B12:

1. Treatment of megaloblastic anemia (Pernicious anemia)
2. Post-gastrectomy state
3. Resection / disease of terminal ileum
4. Subacute combined degeneration of spinal cord

Effects of Vitamin B12 Deficiency:

1. **Megaloblastic anemia**
2. **Neurological abnormalities**
3. **Peripheral neuropathy:**
 - 1. **Subacute combined degeneration of spinal cord (SCD)**
 - 2. Sore tongue (glossitis), angular stomatitis
 - 3. Malabsorption

7. Folic Acid

Sources & Requirement:

- **Source:** Green leafy vegetables, yeast, mushroom, liver, kidney.
- **Daily Requirement:** (50 – 200) µg/day (microgram/day).
- **Increased Requirement:** Increases during pregnancy, lactation, and growing age.

Indications of Folic Acid:

1. **Megaloblastic anemia** (nutritional / folate deficiency)
2. ★ **During pregnancy:** Given periconceptionally — it significantly reduces the risk of **neural tube defects** of the baby.
3. **Malabsorption syndrome** (sprue, celiac disease)
4. **Hemolytic anemia** (due to high bone marrow turnover of RBCs)

Effects of Folic Acid Deficiency:

1. **Megaloblastic anemia**
2. Glossitis, enteritis, diarrhoea
3. ★ **Neural tube defects** — including *spina bifida* and *anencephaly* in the fetus.



1. Hormonal Contraceptives: Classification & Types

Status: Most effective form of birth control. Failure rate of hormonal contraceptives is mainly due to user error / faulty technical compliance.

Classification of Hormonal Contraceptives:

1. Oral contraceptive pill (OCP)
2. Parenteral contraceptives: **DMPA** (Depot medroxyprogesterone acetate) — injectable
3. Implant: **Norplant** — placed subcutaneously via minor incision

Classification of Oral Contraceptive Pills (OCP):

1. **Combined oral contraceptive pill (COC):** Contains both estrogen and progesterone (Efficacy: 99.9%).
2. **Progesterone-only pill (POP) / Mini pill:** Given primarily to lactating mothers.
3. **Emergency contraceptive pill:** Post-coital / morning-after pill.

2. Mechanism of Action of OCP

WRITTEN + VIVA ★ Triple Mechanism of Action:

1. **Inhibition of Ovulation:**
 - **Estrogen** → Inhibits **FSH** release → no follicular growth / maturation.
 - **Progesterone** → Inhibits **LH** release → suppresses pre-ovulatory LH surge → prevents ovulation.
2. **Cervical Mucus Alteration:** **Progesterone** increases the viscosity and thickness of cervical mucus, rendering it impermeable to spermatozoa.
3. **Endometrial Modification:** **Progesterone** alters and thins the endometrial lining, preventing blastocyst implantation.

3. Indications & Non-Contraceptive Benefits of OCP

Indications of OCP:

1. Prevention of conception / Contraception
2. Endometriosis
3. Irregular menstruation
4. Dysfunctional Uterine Bleeding (DUB) — (> 2 weeks & repeated bleeding in married or unmarried females without structural disease)
5. Functional ovarian cysts

SHORT NOTE Non-Contraceptive Benefits:

1. Decreases frequency and severity of dysmenorrhea
2. Relief of premenstrual tension syndrome (PMS)
3. Decreases menstrual blood loss (reduces menorrhagia)
4. Decreases risk of endometrial and ovarian carcinoma
5. Decreases risk of benign fibrocystic disease of ovary & breast
6. Decreases incidence of pelvic inflammatory disease (PID) and ectopic pregnancy

4. Adverse Effects & Contraindications of OCP

WRITTEN + VIVA Adverse Effects of OCP:

Category	Adverse Effects
A. Common A/E	Headache, weight gain, nausea, vertigo (dizziness).
B. Mild A/E	Nausea, mastalgia (breast tenderness), breakthrough bleeding, mild transient bleeding, withdrawal bleeding.
C. Moderate A/E	Breakthrough bleeding, weight gain, chloasma / skin pigmentation, acne, hirsutism, vaginal infection (candidiasis).
D. Severe A/E	Thromboembolism, Myocardial Infarction (MI), Cerebrovascular disease (Stroke), cholestatic jaundice, severe mental depression.

Absolute Contraindications:

1. Previous history of thromboembolic disorders
2. Transient Ischemic Attack (TIA) / Stroke
3. Migraine, SLE (Systemic Lupus Erythematosus)
4. Infective hepatitis / Active liver disease
5. Carcinoma of the breast / Estrogen-dependent tumors
6. Undiagnosed abnormal vaginal bleeding

Relative Contraindications:

1. Family history of thromboembolism
2. Hypertension, Diabetes Mellitus (DM), Heavy smoking
3. Age > 35 years (especially smokers)

⇒ In relative contraindications: switch to injectable progestins, barrier methods, or non-hormonal contraception.

5. Mini Pill & Emergency Contraception

Progesterone-Only Pill (POP) / Mini Pill:

Target Population: Exclusively indicated in lactating / nursing women (during 6 months of exclusive breastfeeding when endogenous estrogen is high and exogenous estrogen can suppress lactation).

Formulations / Drugs: Levonorgestrel, Desogestrel, Norethisterone.

VIVA Advantages:

1. No effect on blood coagulation (no risk of thromboembolism).
2. Can be safely given to women with absolute contraindications to estrogen.
3. Completely safe for nursing mothers (does not impair milk production).

Disadvantages:

1. Less effective than combined oral pills (strict daily timing required).
2. Higher chance of ectopic pregnancy if failure occurs.
3. High incidence of irregular / abnormal uterine bleeding.

Emergency Contraceptive Pill (Morning-After Pill):

• **Intake Window:** Must be taken within **72 hours** of unprotected coitus.

• **Preparations:** High-dose Levonorgestrel, Ethinylestradiol, or Diethylstilbestrol.

6. Long-Acting Hormonal Contraceptives

SHORT NOTE A. Injectable Contraceptives:

- 1. **Medroxyprogesterone acetate** (DMPA / Depo-Provera): 150 mg IM, every 3 months.
- 2. **Norethindrone enanthate** (NET-EN): 200 mg IM, every 2 months.

Advantages: Highly effective, maintains user privacy, no effect on lactation, fertility returns post-discontinuation.

Disadvantages: Requires clinic visits, causes headache, dizziness, and irregular bleeding / amenorrhea.

SHORT NOTE B. Subcutaneous Implants (Norplant):

- **Duration:** Effective for **5 years**.
- **Drug:** **Levonorgestrel** (6 silicone capsules placed sub-dermally).

Advantages: Long-acting, highly reliable, safe in lactation.

Disadvantages: Expensive, requires surgical skill for insertion and removal, irregular bleeding, risk of local site abscess.

7. Formulations of Combined Oral Contraceptives

Pill Formulation	Description & Hormonal Composition
Monophasic Pill	Contains constant, fixed amounts of Estrogen and Progesterone in each active tablet throughout the 21-day cycle.
Biphasic Pill	Contains 2 different combinations: lower progestin dose in the 1st half of cycle to allow endometrial proliferation, followed by a higher progestin dose in the 2nd half.
Triphasic Pill	Contains 3 different combinations of estrogen and progesterone across the cycle, closely mimicking normal physiological hormonal fluctuations.

8. Hormone Replacement Therapy (HRT)

3★ IMPORTANT Definition of HRT:

"The use of estrogen in order to reverse or prevent problems due to loss of ovarian hormone secretion after menopause, whether it is physiological or surgically induced (e.g., bilateral oophorectomy)."

Aims of HRT:

1. **Relief of acute postmenopausal symptoms:** Hot flashes, night sweats, insomnia, lethargy, depression, vaginal dryness / atrophic vaginitis.
2. **Prevention of long-term complications:** Postmenopausal osteoporosis (bone resorption) and coronary artery disease associated with estrogen deficiency.

Preparations Used:

- **Women without uterus (hysterectomized):**
Continuous **Estrogen** alone.
- **Women with intact uterus:** Combined **Estrogen** + **Progesterone** (to prevent endometrial hyperplasia/ carcinoma).

Adverse Effects of HRT:

1. Irregular vaginal bleeding, mastalgia
2. Venous thromboembolism (DVT / PE)
3. Increased risk of endometrial cancer (if unopposed estrogen) and breast cancer

9. Drugs Acting on Uterus & Oxytocic Agents

Classification of Uterine Drugs:

1. **Uterine Stimulants (Oxytocics):** Stimulate uterine contractions and facilitate expulsion of contents.
2. **Uterine Relaxants (Tocolytics):** Inhibit uterine contractions (used to arrest preterm labor).

Classes of Oxytocic Drugs:

- 1. **Oxytocin** (Posterior pituitary peptide hormone)
- 2. Ergot Alkaloids: **Ergometrine**, **Methylergometrine**
- 3. Prostaglandin Analogues: **Misoprostol** (PGE1), **Dinoprostone** (PGE2), **Carboprost** (PGF2 α)

10. Oxytocin

3★ IMPORTANT SHORT NOTE Pharmacological Profile:

- **Nature:** Nonapeptide hormone synthesized in hypothalamus, stored/released from posterior pituitary.
- **Routes of Administration:** Parenteral (IV infusion, IM), Nasal spray.
- **Onset of Action:** 2 minutes • **Half-life ($t_{1/2}$):** 2 – 5 minutes.

Indications of Oxytocin:

1. Induction and augmentation of labour
2. Uterine inertia (hypotonic uterine contractions)
3. Management of incomplete abortion
4. Prevention & control of Postpartum Hemorrhage (PPH)

Adverse Effects of Oxytocin:

1. Fetal distress, neonatal hypoxia, fetal death
2. Uterine rupture (tetanic hyperstimulation)
3. Hypertension, cardiac arrhythmias
4. Water intoxication (ADH-like antidiuretic effect) → hyponatremia, seizures, coma

Contraindications of Oxytocin (Viva Priority):

Fetal distress, prematurity, cephalopelvic disproportion (CPD), abnormal fetal presentation / malpresentation, previous classical cesarean scar (high risk of uterine rupture), and uncontrolled severe maternal hypertension.

Oxytocin, Ergometrine & Prostaglandin

Pharmacological Action of Oxytocin

Slide 1 | Date: 15-08-2026

Pharmacological action :

• Effect on uterus (Oxytocin):

- **In low dose:** It causes full relaxation of uterus in between contraction that mimic normal uterine contraction. Before full term (Labour pain বাড়ানোর জন্য)
- **High dose:** Causes sustained tonic contraction. (Placenta বের করার জন্য)

• Effect on breast :

- Causes immediate contraction of myoepithelium and enhance milk ejection.

• Effect on CVS :

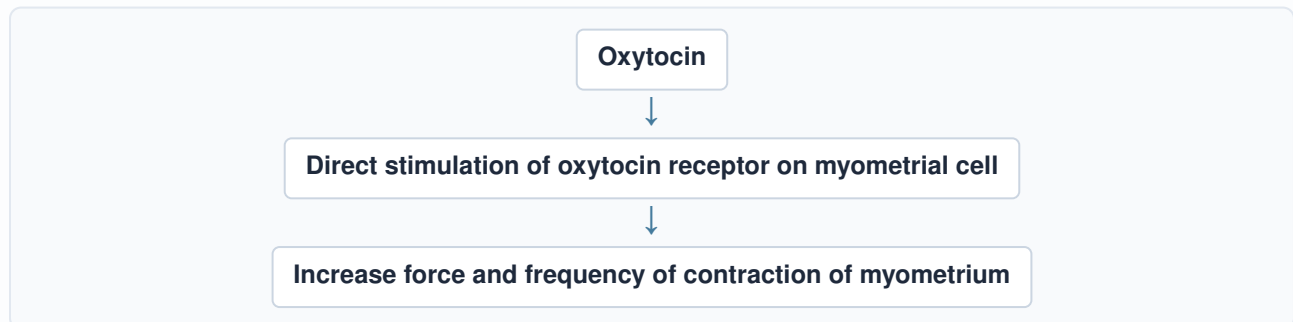
- **In low dose:** Have no effect
- **High dose:** Causes vasodilation and fall of BP.

Important Note: To prevent post partum hemorrhage High dose দেয়। relaxation হবেনা just contraction tonic contraction.

Oxytocin Mechanism of Action & Ergometrine Overview

Slide 2 | Date: 15-08-2026

M/A (Mechanism of Action): Oxytocin



Oxytocin causes contraction by -

1. Activation of voltage gated Ca^{2+} channel.
2. Release of intracellular Ca^{2+}

Oxytocin also increase PG synthesis.

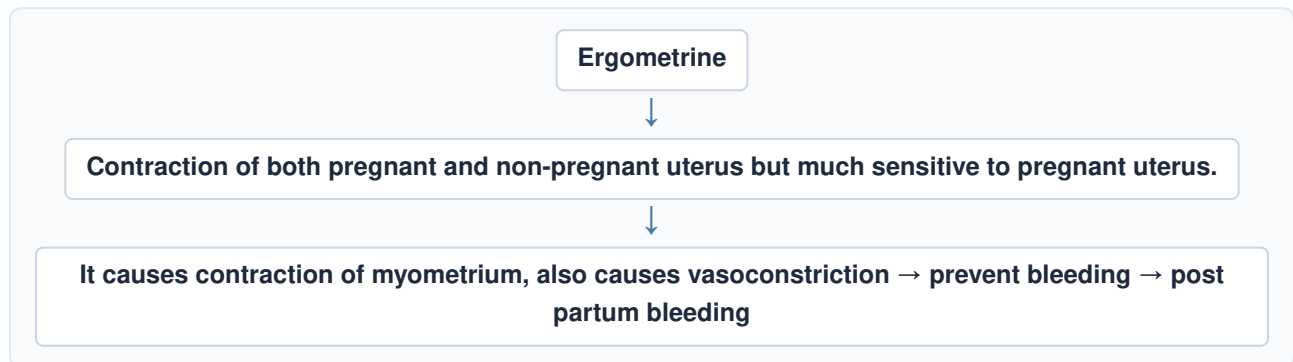
Ergometrine

It is an ergot alkaloid produced by a fungus: *Claviceps purpurea*

- **R/A (Route of Administration):** Parenteral (I/V, I/M), oral
- $T_{1/2}$: 2 hrs.

[V.V.I: Diff btw ergometrine and prostaglandin]

M/A (Mechanism of Action): Ergometrine



Indication :

1. To prevent post partum uterine bleeding

A/E (Adverse Effects):

1. Nausea, vomiting (oral preparation আছে তাই)
2. Increase BP due to increase vasoconstriction
3. Blurring of vision
4. Headache.

Contraindication :

1. 1st and 2nd trimester (Abortion হয়ে যাবে)
2. Hypertension

Prostaglandin

PGE_2 and $PGF_{2\alpha}$ cause contraction of both pregnant and non-pregnant uterus.

- They contract fundus and body of uterus but relaxation of cervix.
- **Chemistry:** It is autacid
- **Onset of action:** Immediate
- $T_{1/2}$: Vary (অনেক আছে PGE_2 , $F_{2..}$)

Pharmacological action :

On gravid uterus, it contracts fundus and body of uterus.

- In early pregnancy, it causes much more contraction than oxytocin (Abortion and product that may causes infection)
- In 2nd and 3rd trimester, it causes strong uterine contraction.
- **On cervix:** It cause ripening and relaxation of cervix

Cervix এর softness বাড়ায়



Strechening এ Help করে → Out of baby

Prostaglandin: Net Effects, Systemic Actions, Routes & Indications**Net effect:**

1. Facilitate labour process
2. Termination of pregnancy.

Systemic Effects :

- **Bronchus:** Causes bronchoconstriction (bronchial asthma তে দিবেনা)
- **CVS:** Vasodilation (PGE_2), Vasoconstriction ($PGF_{2\alpha}$)

R/A (Route of Administration):

1. Parenteral (I/V, I/M)
2. Oral (missabortion)
3. Intravaginal (PPH এ)
4. Intra amniotic

Indication :

1. Termination of pregnancy
2. Induction of labour
3. Prevent post partum hemorrhage.

A/E (Adverse Effects):

1. GIT: Nausea, vomiting, diarrhea
2. Hyperthermia
3. Bronchoconstriction
4. Dysmenorrhea

Contraindication :

1. Asthma
2. Epilepsy (Convulsive disorder)

Parameter / Feature	Oxytocin	Ergometrine	Prostaglandin
1. Chemistry	Peptide hormone	Ergot alkaloid	Autacoid
2. Mode of action	• <i>Moderate dose</i>: Rhythmic uterine contraction • <i>High dose</i>: Sustained tonic uterine contraction	• <i>Moderate dose</i>: Faster contraction, superimposed on tonic contraction • <i>Low dose</i>: Rhythmic uterine contraction and relaxation	On gravid uterus it contracts fundus and body of uterus and relaxation of cervix.
3. R/A	Mainly I/V	I/V, I/M, oral	I/V, I/M, oral, Intravaginal
4. $T_{1/2}$	5 min	2 hrs	Vary
5. Onset of action	2 min	oral: 8 min I/M: 6 min I/V → 1 min	Immediate
6. Indication	Induction of labour	PPH	Abortion
7. Effect on myometrium	Contraction	Contraction	Contraction

**

Prostaglandin Analogue → Collect it.

Essential Drug Concept & Rational Prescribing

ESSENTIAL DRUG CONCEPT

Essential drugs: Drugs that satisfy the health care needs of majority of the populations, they should therefore be available in proper dosage forms, at all the times in adequate amounts, in acceptable quality & at a price that people can afford.

Criteria of Essential drugs:

1. Useful
2. Efficacious
3. Freely available
4. Cost effective
5. Acceptable quality that combat the major health problems of majority people.

Objectives of essential drug project:

1. To make good quality of essential drugs available at minimum cost and in sufficient amount in PHC level.
2. To improve diagnosis & therapy at PHC level.
3. To design & test a strategy for implementation of National Drug policy (NDP).

Why we need EDs:

We need essential drugs because there is:

1. Poor/ little amount of budget in health sector.
2. Unequal distribution of wealth between urban and rural population.
3. Tradition to use medicines instead of emphasizing hygiene and nutrition.
4. Indiscriminate prescribing that often resulting in more harm than good.
5. Proliferation of a wide variety of pharmaceutical products under different name in private sector.
6. Lack of objectives of drug information.

Advantages of EDs:

1. Reduction in the number of pharmaceutical products to be manufactured, imported, stored, analyzed and distributed from amongst the thousands.
2. Improvement of the quality of drug utilization, management, information and monitoring.
3. Stimulation of local pharmaceutical industries.
4. Solution of the PHC problems by providing high priority safe and cost effective drugs.
5. Facilitating the use of drugs more economically and rationally.

Criteria for selection of EDs:

- The EDs should be selected according to the disease profile of the country.
- The selection should be based on evaluations of benefit and safety obtained by clinical trial & epidemiological studies.
- The EDs should be selected by their generic name.
- When several drugs are available with the same therapeutic indication, the drug should be selected by observing the benefit/risk ratio.
- The drugs of fixed dose combination are only acceptable when clinical condition justifies the use of more than one drug or if they show synergistic effects.

Why ED list differ from country to country:

- The prevalence of particular disease.
- Genetic, demographic and environmental factors.
- Financial resources available for health care.
- The level of training of health care personnel.

Essential drug list in BD:

The Essential Drug Company is established in Bangladesh in 1983. Total number of ED in Bangladesh is now 224. These drug have been recommended to be used by various level of health care personnel.

At village level health worker	15
At primary health care level (UHC)	58 (15+58=73)
At secondary level (District hospital)	78 (73+78=151)
At tertiary level (all the above plus)	73 (151+73=224)

P-DRUG CONCEPT

P-drugs: P-drugs are the drugs we have chosen to prescribe regularly and with which we have become familiar.

'P' stands for:

- Personal drug
- Preferred drug
- Priority drug

Criteria for choosing effective groups:

- Efficacy
- Safety
- Suitability
- Cost of the drugs

Steps of choosing P-drugs:

1. **Proper diagnosis:** Ex- Angina pectoris (by partial occlusion of coronary artery).
2. **Specify the therapeutic objectives:** Therapeutic objectives (stopping an attack) can be achieved by reducing myocardial O₂ demand by-
 - Decrease the preload and afterload.
 - Decrease the heart rate and contractility.
3. **Make an inventory of effective groups of drugs:** Nitrates; Beta blocker; Calcium channel blocker
4. **Choose an effective groups according to criteria:**
 - Efficacy
 - Safety
 - Suitability
 - Cost of the drugs
5. **Choose a P-drug:** In general, three active drugs are available for the treatment of an acute attack.
 - Glyceryl trinitrate- Tab/ spray
 - Isosorbide dinitrate- Tab
 - Isosorbide mononitrate- Tab

All these drugs are available in sublingual tablets with a rapid effects.

6. **Finally we can choose a P-drug** by considering the suitability & cost of the drug.

Glyceryl trinitrate (Nitroglycerine)

Dosage form	Sublingual tab (1mg)
Dosage schedule	1 tab stat and 2nd tab given if pain persists.
Duration of treatment	Duration of treatment should be determined by the need or by follow up of the patient.

DRUG ABUSE, DRUG MISUSE & POLYPHARMACY

Drug abuse:

Use of a drug for non medical purpose, usually for altering consciousness is known as drug abuse.

Eg: Morphine, when it is used for euphoria.

Drug misuse:

Use of drug for wrong indication, in the wrong dosage or for a too long period is called drug misuse.

- Use of antibiotics for viral fever/ infection.
- Use of parenteral preparation of drugs when it is not required.

Polypharmacy:

Administration of multiple drugs (more than 2 drugs) simultaneously is known as polypharmacy.

For many people, appropriate polypharmacy may extend life expectancy and improve quality of life.

1. **In Hypertension**, when BP is not controlled by monotherapy, then it is treated by polypharmacy.

Eg: Amlodipine+ Atenolol+ Hydrochlorothiazide

2. **In the first year after myocardial infarction:**

Eg: Statin ACEI + Beta blocker + Aspirin + Antidepressant should be given.

In TB, we use many drugs at a time. But it is not polypharmacy.

Though polypharmacy can be beneficial but sometimes it may be harmful. It may concern with increased adverse effects, drug reaction and higher costs.

RATIONAL USE OF DRUGS AND RATIONAL PRESCRIBING

According to WHO:

Rational use of drugs requires that patients receive medications appropriate to their clinical needs, in doses that meet their own requirements, for an adequate period of time and at lowest cost to them and their community.

Rational use of drug is based on Rule of Right: Right drug is given to the Right patient at Right time with Right doses.

They should also fulfill the following criteria (SANE):

- | | |
|-----------------|------------|
| • Safety | • Need |
| • Affordability | • Efficacy |

Criteria for rational prescribing:

1. **Appropriate diagnosis:** On the basis of clinical feature and laboratory investigations.
2. **Appropriate indication:** Drug should be prescribed only when it is necessary. Eg: Most of the diarrhea in children is due to viral (Rota virus) cause and here antibiotic is not needed.
3. **Appropriate drug:** The selection of drug is based on efficacy, safety, suitability, cost considerations and easy availability.
4. **Appropriate patient:** Prescriber should consider patient's sensitivity to drug. Eg: During pregnancy and lactation, some drugs are contraindicated.
5. **Appropriate administration:** Proper dose, route, dosage schedule and duration should be maintained.
6. **Appropriate information:** The patient should be provided with relevant, accurate, important, and clear information that are prescribed.
7. **Appropriate monitoring:** The anticipated and unexpected effects of medication should be appropriately monitored.

Common patterns of irrational prescribing:

1. Use of drug when no drug is indicated. Eg: Use of antibiotic in viral infection.
2. Use of wrong drug in specific condition. Eg: Tetracycline in childhood diarrhea.
3. Use of drug with doubtful and unproven efficacy. Eg: Use of Loperamide in acute diarrhea.
4. Failure to provide available, safe and effective drug. Eg: Failure to prescribe ORS in acute diarrhea.
5. Prescribing new costly drug when an alternative cheaper drug of equal efficacy is available.
6. Use of correct drugs with incorrect administration, dose and duration, Eg: Use of I/V Metronidazole when oral preparation would be appropriate.
7. Indiscriminate use of antibiotics and polypharmacy.

Factor affecting the irrational use of drug:

A. Factor related to the patient:

1. Lack of proper information.
2. Patient's demand.
3. Easy to purchase any drug without prescription.
4. Misleading beliefs, "For every ill, there is a pill."

C. Dispenser:

1. Dispenser fails to supply the drug properly.

E. Legal authorities:

1. Non essential drugs are available in the market.
2. There is no restriction for prescribing vital drugs and antibiotics by informal prescribers.
3. Regulation enforcements is very poor in developing countries.

Impact of irrational prescribing:

1. The reduction in the quality of drug therapy leading to increased morbidity and mortality.
2. Waste of financial resources leading to reduced utilization of other vital drugs.
3. Increased risk of unwanted effects.
4. Psychological impact is that patient come to believe "there is a pill for every ill."

B. Factor related to the prescriber:

1. Lack of continuous medical education and training for updating the knowledge.
2. Insufficient staffing pattern of the hospital.
3. Sometimes prescribers are instructed to prescribe limited number of drugs.
4. Misleading belief about drug efficacy.

D. Factors related to the drug supplier

1. Unreliable supplier
2. Artificial storage of essential drugs
3. Supplying expired drugs.

Prevention of irrational prescribing:

To prevent irrational prescribing the following measures should be taken:

1. Making correct diagnosis.
2. Limiting the number of drugs.
3. Encouraging the availability of essential drugs.
4. Providing adequate training, drug information & standard treatment guideline to the prescribers and incorporating the concept of essential drugs through continuous education.
5. Teaching of rational prescribing into the curriculum of medicine, pharmacy, dentistry and nursing.
6. Finally providing effective public education to the consumers and on public.

FLUOROQUINOLONES

Chemistry, Limitations of Older Quinolones, Advantages & Spectrum Grouping

1. CLASSIFICATION & CHEMICAL STRUCTURE

1. Non-Fluorinated Quinolone • Prototype Drug: Nalidixic acid • Older synthetic agent lacking fluorine substitution. • Narrow spectrum with limited systemic antibacterial efficacy.	2. Fluorinated Quinolone (Fluoroquinolone) • Structural Feature: A Fluorine atom is attached to the 6th position of the basic quinolone ring structure. • Significantly enhances potency, tissue penetration, and spectrum.
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2. LIMITATIONS OF OLDER QUINOLONES VS. ADVANTAGES OF FLUOROQUINOLONES

LIMITATIONS OF OLDER QUINOLONES (NALIDIXIC ACID)	ADVANTAGES OF FLUOROQUINOLONES
1. Narrow spectrum: Predominantly active against Gram-negative bacteria only.	1. Expanded spectrum: Broad-spectrum bactericidal action (both Gram-negative & Gram-positive).
2. Low potency.	2. High potency.
3. Modest blood & tissue levels: Poor systemic tissue distribution.	3. Better tissue penetration: High intracellular and tissue concentrations.
4. High frequency of bacterial resistance.	4. Slow development of bacterial resistance (No cross-resistance).
5. High MIC: Minimum Inhibitory Concentration for most bacteria is ~20 µg/mL.	5. Low MIC: Minimum Inhibitory Concentration is low, between 0.5 – 6 µg/mL.

Viva Pearl (Antimicrobial Spectrum Coverage): While Cephalosporins broaden from Gram-positive to Gram-negative (+ve → -ve) across generations, Fluoroquinolones progress from Gram-negative to Gram-positive (-ve → +ve).

3. SPECTRUM COVERAGE & GENERATIONAL GROUPING

GROUP / CLASS	SPECTRUM & DISTINCTIVE FEATURES	REPRESENTATIVE DRUGS
1st Group	Least active against Gram-positive and Gram-negative organisms; limited primarily to urinary tract infections.	Norfloxacin
2nd Group	• Gram-negative: Excellent activity. • Gram-positive: Moderate to good activity. • Widely used clinical workhorses across systemic infections.	Ciprofloxacin , Levofloxacin , Lomefloxacin , Nadifloxacin , Pefloxacin , Ofloxacin
3rd Group	• Gram-positive: Significantly improved/enhanced activity. • Gram-negative: Retains excellent activity of 2nd group.	Gatifloxacin , Moxifloxacin , Gemifloxacin
4. Respiratory Fluoroquinolones	• Rationale & Naming: Specifically indicated in lung / respiratory tract diseases. • Enhanced Gram-positive potency (e.g., <i>S. pneumoniae</i>). • Highly active against Atypical pneumonia agents (e.g., <i>Chlamydia</i>, <i>Mycoplasma</i>).	Levofloxacin , Moxifloxacin , Gemifloxacin , Gatifloxacin

4. MECHANISM OF ACTION (MOA)

Fluoroquinolones block bacterial DNA synthesis → Produces rapid Bactericidal effect

- **Inhibition of Topoisomerase II (DNA Gyrase):** Prevents relaxation of positively supercoiled DNA, which is required for bacterial DNA transcription and replication.
- **Inhibition of Topoisomerase IV:** Interferes with the separation (decatenation) of replicated chromosomal DNA into respective daughter cells during cell division.

5. PHARMACOKINETICS & INDICATIONS

Route (RA)	Oral, Parenteral (IV-pre-op/systemic)	Bioavailability	~95% (High oral bioavailability)
Metabolism	Hepatic (Liver) - enzyme inhibitor	Excretion	Dual route: Renal & Bile

Indications of Respiratory Fluoroquinolones:

1. Community-Acquired Pneumonia (CAP)
2. Bronchitis & Sinusitis
3. Skin and Soft Tissue Infections
4. MDR-TB: Especially **Levofloxacin**
5. Chlamydial Urethritis

Must-to-Know Indications of **Ciprofloxacin** :

1. Urinary Tract Infections (UTI)
2. Respiratory Tract Infections: Upper & lower (sinusitis, pharyngitis, bronchitis)
3. Enteric Fever (Typhoid): **DRUG OF CHOICE**
4. Skin & Soft Tissue Infections
5. Bacterial Diarrhea & Dysentery
6. Intra-Abdominal Infections: Mixed with **Metronidazole** (anaerobic coverage)
7. PID & STDs (Pelvic inflammatory disease)
8. TB & Atypical Mycobacterial Infections
9. Anthrax (*Bacillus anthracis*)

6. ADVERSE EFFECTS & CONTRAINDICATIONS

Adverse Effects (A/E)		CONTRAINDICATIONS & RATIONALE	
Hypersensitivity	Fever, skin rash, urticaria, itching, angioedema.	1. Known hypersensitivity to fluoroquinolones.	
GIT	Anorexia, nausea, vomiting, abdominal discomfort, diarrhea.	2. Children below 18 years: Risk of cartilage damage/arthritis in weight-bearing joints.	
Arthropathy	Damage to growing cartilage; not suitable for children <18 yrs (reversible).	3. Pregnancy: Inhibits DNA synthesis; fetal chondrotoxicity risk.	
Tendinitis	Inflammation of tendon & Achilles tendon rupture (in adults).	4. Athletes: High risk of tendinitis and tendon rupture.	
CNS	Headache, dizziness, confusion, insomnia, convulsions.	5. Co-administration with Antacids & Theophylline:	
Hepatic	Abnormal Liver Function Tests (LFT).	◦ Antacids: Chelation impairs FQ absorption.	
Skin	Photosensitivity reactions.	◦ Theophylline: Enzyme inhibition leads to theophylline accumulation and toxicity.	
Cardiac	QT Prolongation: Causes cardiac arrhythmias (Gatifloxacin , Gemifloxacin , Moxifloxacin , Levofloxacin).		
Drug Interaction	Hepatic microsomal enzyme inhibitor.		

1. Anti-Amoebic Drugs

Definition: These are the drugs used to treat infection caused by the anaerobic protozoa *Entamoeba histolytica*.

Forms of *Entamoeba histolytica*:

Entamoeba histolytica exists in **2 forms**:

- **(i) Cyst (Infective form):** Asymptomatic, carrier state (*carries infection from one host to another but does not produce symptoms*).
- **(ii) Trophozoite:** Non-infective form, symptomatic.

Invasive form leading to:

1. Amoebic dysentery
2. Amoebic liver abscess
3. Hepatic amoebiasis

Classification of Amoebicidal Drugs:

(I) Tissue Amoebicide:

(Effective in trophozoite state)

1. **Metronidazole**
2. **Tinidazole**
3. **Emetine hydrochloride**
4. **Dehydroemetine**
5. **Chloroquine**

(II) Luminal Amoebicide:

(Acts in intestinal lumen)

1. **Diloxanide furoate**
2. **Paromomycin**
3. **Tetracycline**
4. **Erythromycin**

2. Metronidazole

- **Physical Characteristics:** Yellow colour and metallic taste.
- **Nature:** It is a prodrug; prototype drug is **nitro-imidazole**.
- **Spectrum:** Broad spectrum anti-protozoal drug.
- **Pharmacokinetics:** Oral **metronidazole** is readily absorbed and permeates all tissues by simple diffusion.
- **Metabolism & Excretion:** Metabolized in liver and excreted by kidney.
- **Action:** Kills trophozoite form; **bactericidal** in nature.
- **Dose:**
 - **Oral:** 400 mg (*also available as 500 mg*)
 - **IV:** 500 mg

Mechanism of Action:

Metronidazole enters into protozoa and sensitive protozoal cell



After entering the cell by diffusion, its **nitro group** is reduced into a **highly reactive nitro radical** by **ferredoxin**



Combines with **DNA** and causes **impairment of DNA function**



Resulting in death of protozoa

Indications:

1. Amoebiasis
2. Amoebic liver abscess
3. Giardiasis
4. Urogenital trichomoniasis
5. Preparation of bowel for surgery (*with **cephalosporin***)
6. Gingivitis, dental infection
7. Leg ulcer, decubitus ulcer
8. Genital infection
9. Sepsis / intra-abdominal infection
10. Brain abscess
11. PUD (to treat *H. pylori* peptic ulcer) → Triple therapy

Adverse Effects:

- **(i) GIT:** Anorexia, nausea, metallic taste in mouth, abdominal discomfort.
- **(ii) CNS:** Dizziness, vertigo, ataxia, numbness.
- **(iii) Prolonged use:** May cause peripheral neuropathy.
- **(iv) Oral:** Glossitis.
- **(v) Drug interaction:** Hepatic enzyme inhibitor — increases toxicity with **warfarin**, **cimetidine**, etc.

Contraindications:

1. **Neurological disorder**
2. **Blood dyscrasias**
3. **1st trimester pregnancy**

Management of Hepatic Amoebiasis:

- **Regimen 1:**
 - Tab. **Metronidazole** (800 mg): TDS for 10 days
 - Tab. **Diloxanide furoate** (500 mg): TDS for 10 days
- **Regimen 2 (Alternative):**
 - Tab. **Chloroquine** (300 mg): 2 tab BD for 2 days → 1 tab BD for 14 – 21 days
 - Tab. **Diloxanide furoate** (500 mg): TDS for 20 days

3. Anti-Helminthic Drugs

Definition: Anti-helminthic drugs are the drugs or agents which are used to eradicate or reduce the number of helminthic parasite in the intestinal tract or tissue in the body.

Classification:

- (i) For Nematodes: e.g., **Albendazole**, **Mebendazole**, **Pyrantel pamoate**, **Levamisole**.
- (ii) For Cestoda: **Niclosamide**.
- (iii) For Trematode: **Praziquantel**.

Albendazole

Dose: 400 mg

Mechanism of Action:

Albendazole



Inhibits microtubule synthesis in nematode



Irreversibly blocks glucose uptake by helminth



Depleting glycogen stores



Decreases ATP synthesis of worm



Parasite is immobilized and died

Indications:

1. Ascariasis
2. Trichuriasis
3. Hookworm
4. Cysticercosis
5. Pin worm infection
6. Strongyloidiasis
7. Cutaneous larvae migrans
8. Hydatid disease

Adverse Effects:

- (i) **GIT:** Anorexia, nausea, vomiting, abdominal cramp, diarrhoea
- (ii) **CNS:** Headache, dizziness
- (iii) Alopecia
- (iv) Pancytopenia

Contraindications:

1. **Hypersensitivity**
2. **Pregnancy**
3. **Children below 2 years**

Filariasis Treatment:

- (i) **Diethylcarbamazine (2 – 3 mg/kg):** TDS for 2 – 3 weeks in Bancroftian filariasis.
- (ii) **Diethylcarbamazine (6 mg/kg):** BD for Tropical pulmonary eosinophilia.
- (iii) **Mebendazole:** Dose — 100 mg.

4. Antifungal Drugs

Overview: Fungal infections are called **mycosis**. Candidiasis occurs in the mucous membrane of skin, mouth, vagina, alimentary tract.

Types of Fungal Infection:

1. **Dermatophytosis / Ring worm:** Affects skin, hair, nail.
2. **Tinea pedis:** Foot
3. **Tinea corporis:** Trunk, arm, leg
4. **Tinea capitis:** Scalp
5. **Tinea cruris:** Groin
6. **Tinea onychomycosis:** Nail

Classification of Antifungal Drugs:

1. According to Mechanism of Action:

- (i) Drugs that disrupt fungal cell membrane: e.g., **Amphotericin B**, **Nystatin**, **Imidazole**, **Triazole**, **Terbinafine**.
- (ii) Drugs that interrupt fungal cell wall: e.g., **Caspofungin**.
- (iii) Drugs that inhibit mitosis: e.g., **Griseofulvin**.
- (iv) Drugs that inhibit DNA and RNA synthesis: e.g., **Flucytosine**.

2. According to Disease:

- 1. Subcutaneous and Systemic Mycoses: e.g., **Amphotericin B**, **Caspofungin**, **Fluconazole**, **Itraconazole**, **Ketoconazole**, **Voriconazole**, etc.
- 2. Cutaneous Mycoses: e.g., **Nystatin**, **Terbinafine**, **Griseofulvin**, **Clotrimazole**.

5. Azole Antifungals

Definition: Azoles are a group of synthetic antifungal agents with broad spectrum activity.

Classification of Azoles:

(I) Imidazole:

(Mainly topical formulation)

1. **Clotrimazole**
2. **Miconazole**
3. **Ketoconazole** (topical + systemic)
4. **Econazole**

(II) Triazole:

(Systemic formulation)

- **Fluconazole**
- **Itraconazole**
- **Voriconazole**

Mechanism of Action:

- Azoles inhibit fungal **cytochrome P₄₅₀ enzyme**, responsible for conversion of **lanosterol to ergosterol** (main sterol in fungal cell membrane).
- Inhibit transformation of candidal yeast cells into hyphae.
- **Ketoconazole**: Also interferes with fungal oxidative enzymes → lethal accumulation of **hydrogen peroxide**.

Adverse Effects:

* Note: **Ketoconazole** causes topical + systemic + endocrine effects; **Fluconazole** causes only systemic adverse effects.

- **Topical Azoles**: Tingling sensation, pruritus, erythema, local irritation.
- **Systemic Azoles**:
 - **GIT**: Nausea, vomiting (oral).
 - **Liver**: Increased transaminases.
- **Endocrine (Ketoconazole)**: Inhibits steroid synthesis → Gynaecomastia, decreased libido, impotence, menstrual disturbance.
- **Drug Interaction**: Inhibits CYP450 → toxicity of **warfarin**, **phenytoin**, **cyclosporin**.

Advantages of Fluconazole over Ketoconazole:

1. More H₂O soluble
2. Wide range of activity
3. High oral bioavailability (not hampered by food/pH)
4. Good CSF penetration
5. Less frequency of dosing (T_{1/2}: 22 – 31 hrs)
1. Wider therapeutic index
2. Better selectivity to fungal CYP450 (less adverse effects)
3. No endocrine side effects
4. Can be used both orally and parenterally

6. Specific Treatment Regimens

Onychomycosis (Nail Infection):

- (i) **Terbinafine**: 250 mg daily — **12 weeks**, OR
- (ii) **Itraconazole**: 250 mg — **3 months**, OR
- (iii) **Griseofulvin**: 250 mg — **4 – 6 months**

Candidiasis Treatment:

- **Oral**: Ointment → **Miconazole** OR **Fluconazole**
- **Vaginal**: **Nystatin**, **Fluconazole**

Thyroid Hormone & Anti Thyroid Drugs

Thyroid Function & Hormones

The primary function of thyroid gland is production of iodine containing thyroid hormone.

1. tri iodothyronine (T_3)
2. Thyroxine (T_4)
3. Calcitonin

Thyroid Preparation –

1. Levothyroxine
2. Liothyronine (3–4 more potent)

T_4 is converted into T_3 intracellularly. So, administered of T_4 produce both hormone.

Abnormality of Thyroid Function:

1. Hyperthyroidism (thyrotoxicosis)
2. Hypothyroidism

Anti-Thyroid Drugs –

1. Thioamides
 - Carbimazole
 - Methimazole
 - Propylthiouracil
2. Radioactive iodine I^{131}
3. Iodide
4. β blocker – propranolol
5. Anion inhibitor – perchlorate, thiocyanate

Role of β Blocker in Thyrotoxicosis :

1. $\uparrow$ sympathetic activity in hyperthyroidism
2. $\downarrow$ anxiety, irritability, palpitation, tremor
3. $\uparrow$ hormone synthesis

So, it is used to relief thyrotoxic symptoms.

Carbimazole – It is a prodrug. It is converted into **methimazole**.

M/A :

- Prevent hormone synthesis by inhibiting thyroid peroxidase-catalyzed reaction & blocking iodine organification.
- Blocking coupling of iodotyrosine to form T_3 & T_4 .

Adverse Effect :

1. GIT: Nausea, GIT upset
2. Fever, maculopapular rash
3. Vasculitis
4. Arthralgia
5. Lymphadenopathy
6. Hepatitis
7. Agranulocytosis (severe case)
8. Teratogenesis

→ **Propylthiouracil** has no teratogenesis effect.

Anti Cancer drug

Cancer

An abnormal growth of cells which tends to proliferate in an uncontrolled way and in some cases it can be metastasize.

Cancer can involve any tissue of the body & have different forms.

Principal modalities of cancer treatment

1. Surgery
2. Radio therapy
3. Chemotherapy
4. Endocrine therapy
5. Immuno therapy
6. Biological therapy.

Cell cycle & Phase of cell cycle

Cell cycle: Cell cycle means sequence of events that takes place within a cell for its division.

Phase of cell cycle –

- **M phase mitosis** – division into two daughter cell.
- **S** → phase for DNA synthesis
- **G₁** → phase for preparation of DNA synthesis (resting phase)
- **G₂** → phase for preparation for mitosis
- **G₀** → quiescent (resting phase)

Classification of anti-cancer drug :

A. Cell cycle specific drugs :

kills only that those are actively cycling.

1. **Antimetabolites** – methotrexate , 5-Fluorouracil , mercaptopurine .
2. **Plant alkaloids** – vincristine , vinblastine , etoposide
3. **Anti-tumor antibiotic** – Bleomycin

B. Cell cycle non-specific drug –

kills cells whether cells are resting or not.

1. **Alkylating agent** : cyclophosphamide .
2. **Antibiotics** : Daunorubicin , doxorubicin , mitomycin
3. **Platinum analogue** – Cisplatin .
4. **Hormones** – Hydrocortisone , testosterone , diethylstilbestrol .
5. **Selective estrogen receptor modulator (SERM)** : Tamoxifen , clomifen

Adverse effect

1. **Bone marrow suppression** : Agranulocytosis, thrombocytopenia, aplastic anemia.
2. **Lymphoreticular tissue** – lymphocytopenia
3. **GIT** : anorexia, nausea, vomiting, diarrhea, stomatitis.
4. **Alopecia**
5. **Endocrine** : oligospermia, amenorrhea
6. **Fetus** : abortion, fetal death, teratogenesis.
7. **Neuropathy, cardiomyopathy.**